

**Lecture 3-2: atomic shells and classical
models of chemical bonding-1**

Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**

1. **Representations of molecules**

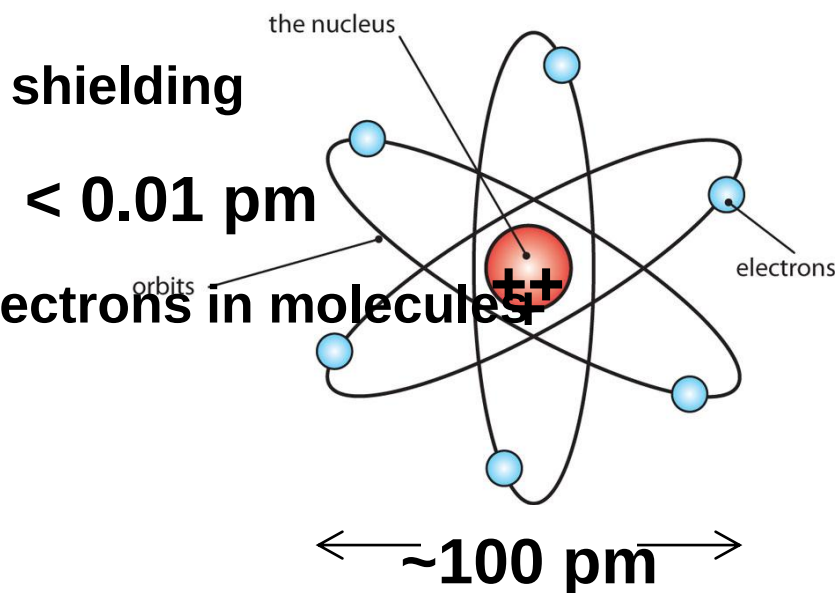
2. **The periodic table**

3. **Forces and potential energy in atoms**

4. **Ionization energies, the shell model of the atom, and shielding**

5. **Electron affinity**

6. **Electronegativity: the tendency of atoms to attract electrons in molecules**



Molecules

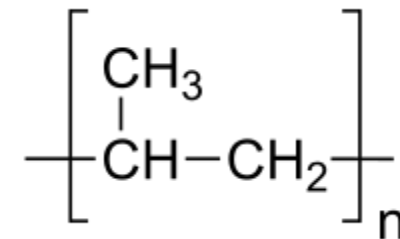
- ❑ **Molecule: a collection of atoms bonded together, with the elements in fixed proportions and with a (generally) well-defined three-dimensional structure.**

- **Diatomic molecule**

$\text{N}_2, \text{O}_2, \text{Cl}_2, \text{Br}_2, \text{I}_2, \text{CO}$

- **Polyatomic molecules**

$\text{H}_2\text{O}, \text{NH}_3, \text{CH}_4$



Molecular Formulas

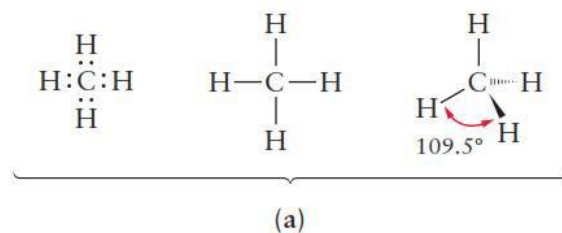
- ❑ Molecule: a collection of atoms bonded together, with the elements in fixed proportions and with a (generally) well-defined three-dimensional structure.

	Hydrogen	Water	Ammonia	Methane
Molecular formula	H_2	H_2O	NH_3	CH_4
Structural formula	$\text{H}-\text{H}$	$\text{H}-\text{O}-\text{H}$	$\begin{array}{c} \text{H}-\text{N}-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array}$

- ❑ Held together by chemical forces.

Molecular representations

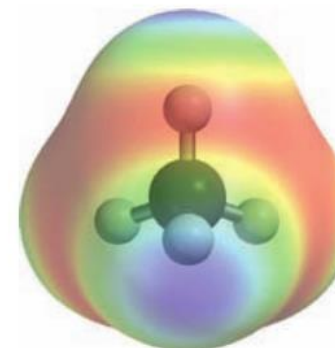
Methane



(b)

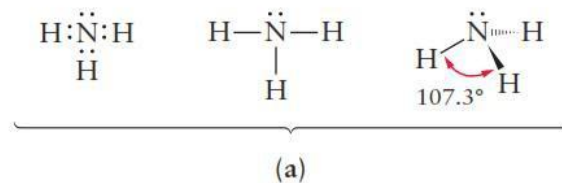


(c)

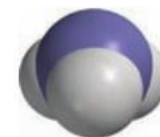


(d)

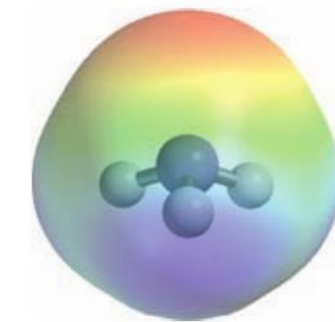
Ammonia



(b)

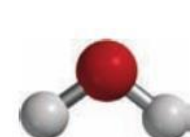
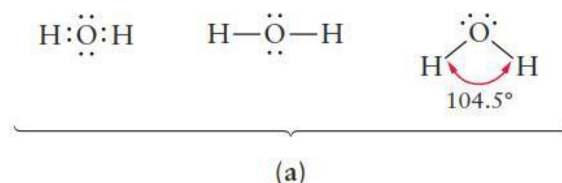


(c)

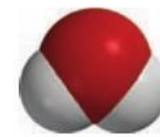


(d)

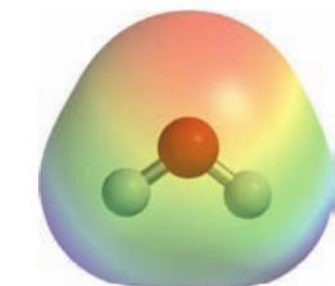
Water



(b)



(c)



(d)

Lewis dot

**line
structures**

**line
angle**

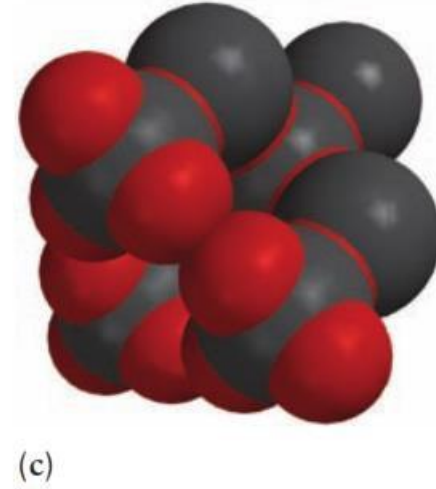
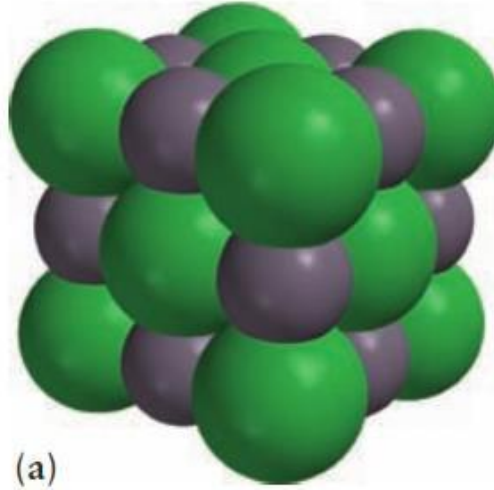
**ball and
stick**

**space
filling**

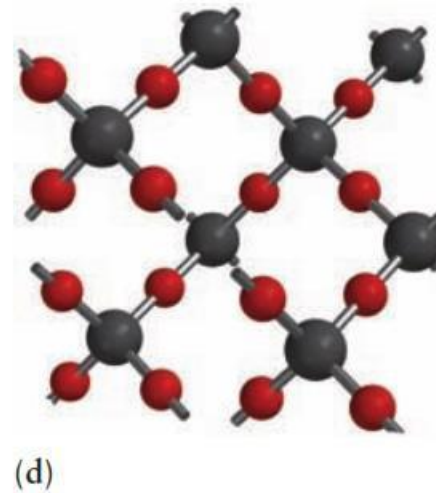
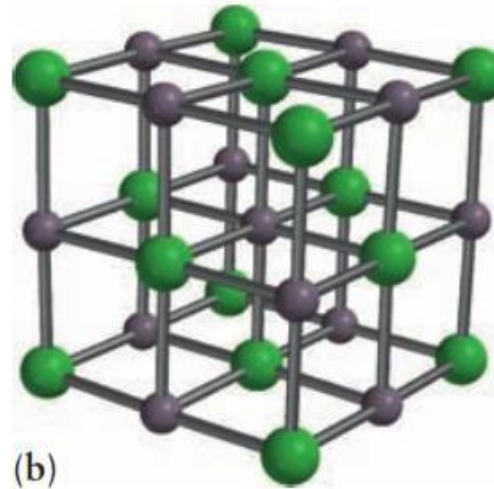
**electrostatic
potential
energy**

Crystalline forms

Space filling



ball and stick



NaCl

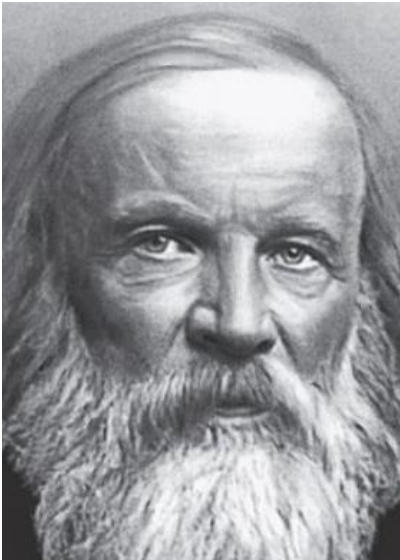
SiO₂

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 1. Representations of molecules
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Mendeleev's Periodic Table

Periodic law: the chemical properties of the elements are periodic functions of the atomic number Z.



Dmitri
Mendeleev
(1834–1907)

Reihen	Gruppe I. — R'O	Gruppe II. — RO	Gruppe III. — R'O ³	Gruppe IV. RH ⁴ RO ²	Gruppe V. RH ³ R'O ³	Gruppe VI. RH ³ RO ³	Gruppe VII. RH R'O ¹	Gruppe VIII. — RO ⁴
1	II=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	{Au=199}	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —

(1871)

The Periodic Table

Representative elements
主族元素

Transition-metal elements
过渡金属元素

Groups (族): arranged vertically

Periods (周期): arranged horizontally

												主族元素							
I												III	IV	V	VI	VII	VIII		
1 H 1766	2 II												13 III	14 IV	15 V	16 VI	17 VII	18 VIII	
3 Li 1817	4 Be 1798												5 B 1808	6 C 1808	7 N 1772	8 O 1772	9 F 1887	10 Ne 1898	
TRANSITION ELEMENTS																			
11 Na 1807	12 Mg 1756	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIII	9 VIII	10 VIII	11 IB	12 IIB	13 Al 1827	14 Si 1823	15 P 1669	16 S 1669	17 Cl 1774	18 Ar 1894		
19 K 1807	20 Ca 1808	21 Sc 1879	22 Ti 1791	23 V 1830	24 Cr 1797	25 Mn 1774	26 Fe 1735	27 Co 1735	28 Ni 1751	29 Cu 1746	30 Zn 1746	31 Ga 1875	32 Ge 1886	33 As 1817	34 Se 1817	35 Br 1826	36 Kr 1898		
37 Rb 1861	38 Sr 1790	39 Y 1794	40 Zr 1789	41 Nb 1801	42 Mo 1778	43 Tc 1937	44 Ru 1844	45 Rh 1803	46 Pd 1803	47 Ag 1817	48 Cd 1817	49 In 1863	50 Sn 1863	51 Sb 1863	52 Te 1782	53 I 1811	54 Xe 1898		
55 Cs 1860	56 Ba 1808	71 Lu 1907	72 Hf 1923	73 Ta 1802	74 W 1781	75 Re 1925	76 Os 1803	77 Ir 1803	78 Pt 1735	79 Au 1735	80 Hg 1735	81 Tl 1861	82 Pb 1861	83 Bi 1861	84 Po 1898	85 At 1940	86 Rn 1900		
87 Fr 1939	88 Ra 1898	103 Lr 1961	104 Rf 1965	105 Db 1970	106 Sg 1976	107 Bh 1976	108 Hs 1984	109 Mt 1982	110 Uun 1994	111 Uuu 1995	112 Uub 1996	114 Uuq 1999							
LANTHANIDES																			
		57 La 1839	58 Ce 1803	59 Pr 1885	60 Nd 1843	61 Pm 1947	62 Sm 1879	63 Eu 1896	64 Gd 1880	65 Tb 1843	66 Dy 1886	67 Ho 1879	68 Er 1843	69 Tm 1879	70 Yb 1907				
		ACTINIDES																	
		89 Ac 1899	90 Th 1828	91 Pa 1917	92 U 1789	93 Np 1940	94 Pu 1940	95 Am 1945	96 Cm 1944	97 Bk 1950	98 Cf 1950	99 Es 1952	100 Fm 1953	101 Md 1955	102 No 1958				

The Periodic Table

- **Metals** and **nonmetals**: distinguished by the presence (or absence) of a characteristic metallic luster, good (or poor) ability to conduct electricity and heat, and malleability (or brittleness).
- **semimetals** (boron, silicon, germanium, arsenic, antimony, and tellurium) resemble metals in some respects and nonmetals in others.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
															Pnictogens	Chalcogens	Halogens		
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight C Solid Hg Liquid H Gas Rf Unknown Metals Alkali metals Alkaline earth metals Lanthanoids Actinoids Transition metals Post-transition metals Metalloids Nonmetals Reactive nonmetals Noble gases																2 He Helium 4.0026	
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122																	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305																	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798	
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29	
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)	
7	87 Fr Francium (223)	88 Ra Radium (226)	89–103	104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)	
For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.																			
	57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97				
	89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)				

The Periodic Table

- Group I: alkali metals. Relatively soft metals with low melting points. LiCl and NaCl
- Group II: alkaline-earth metals. MgCl₂ and CaCl₂
- Group VI: chalcogens. CaO and BaS; Li₂O and Na₂S
- Group VII: halogens. LiF and LiCl

- Group VIII: **noble gases** (He, Ne, Ar, Kr, Xe, Rn), also called the **inert gases** because of their relative inertness toward chemical combination. (monatomic)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
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Background Music
Gioacchino Rossini *La gazza ladra* Ouverture

锂盐（英语：Lithium salts）为含**锂离子**的**离子化合物**，临床上通常指**碳酸锂**。锂盐在医学上可作为一种**精神科**药物，用于**双相障碍**与其他**抗抑郁药**治疗无效的**重性抑郁障碍**^[2]。锂能够降低这些疾病患者的症状发生，进而减少患者**自杀**的机会。^[3]，经口服后由胃肠吸收，而经肾脏由尿代谢^[2]。

锂盐的常见副作用有**频尿**、手抖、流涎（sialorrhea）、钾离子缺乏等^[2]。严重者有**甲状腺机能低下症**、**尿崩症**与**锂中毒**^[2]。因此建议监测其血中浓度，降低中毒风险^[2]。如果血液中锂浓度过高，可能出现腹泻、呕吐、协调不良、嗜睡、**耳鸣**^[2]。**孕妇**使用锂化合物可能造成胎儿发育问题^[2]。**哺乳期**妇女服用则似无问题^[4]。锂化合物属一种**心境稳定剂**^[2]。这种药物的作用机转尚不明确。^[2]

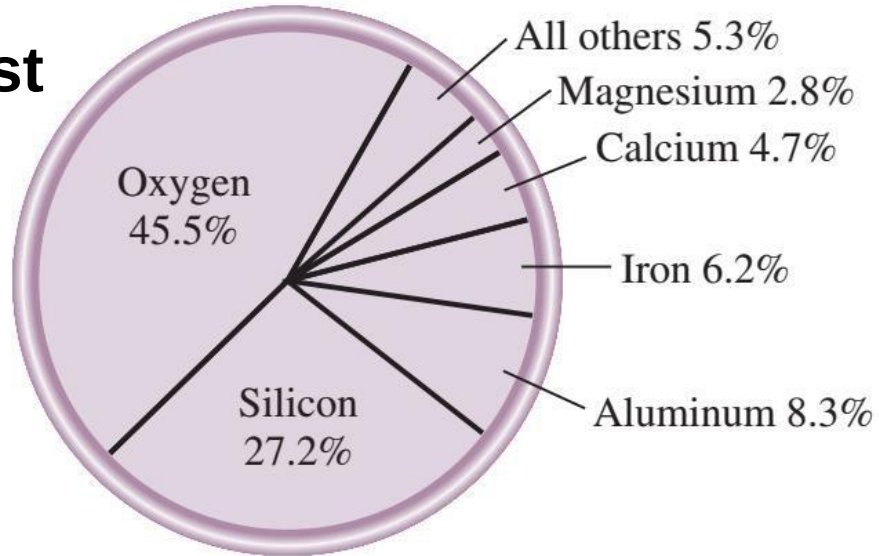


*图片仅供参考，产品以实物为准

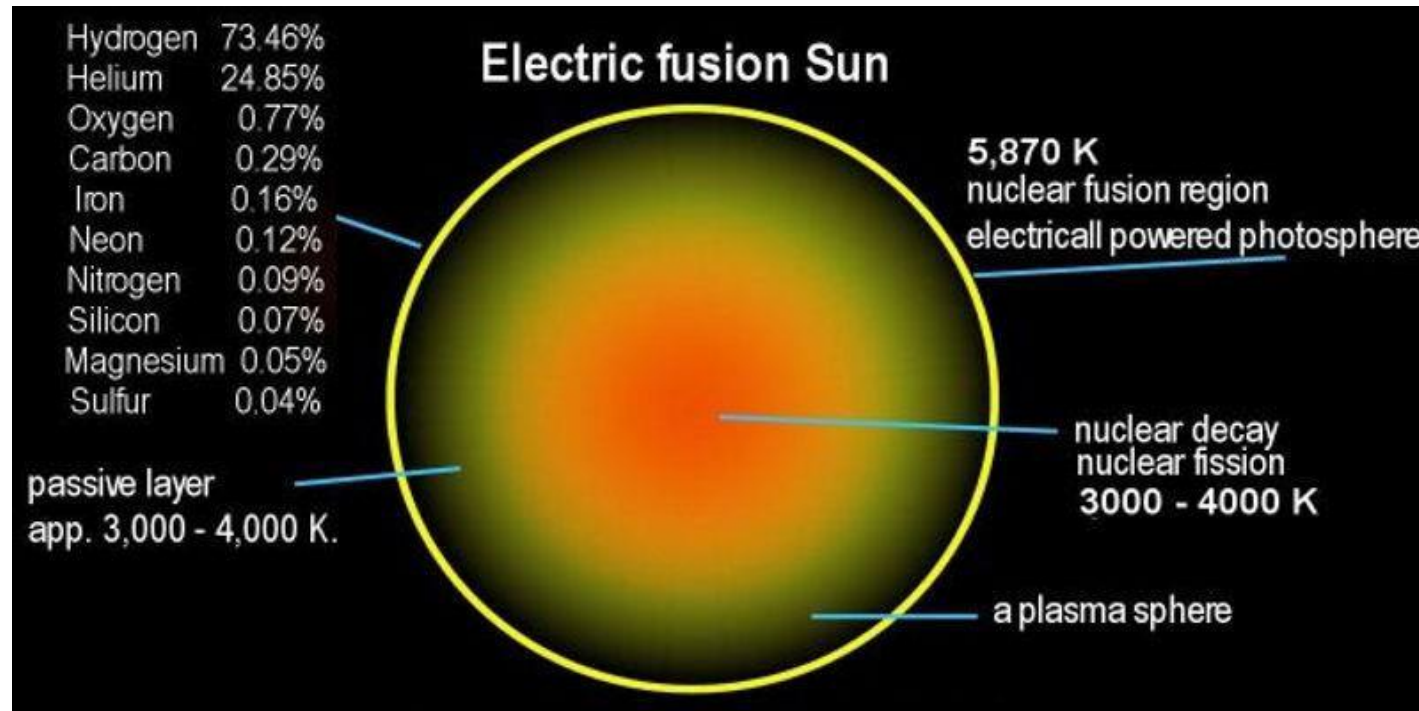
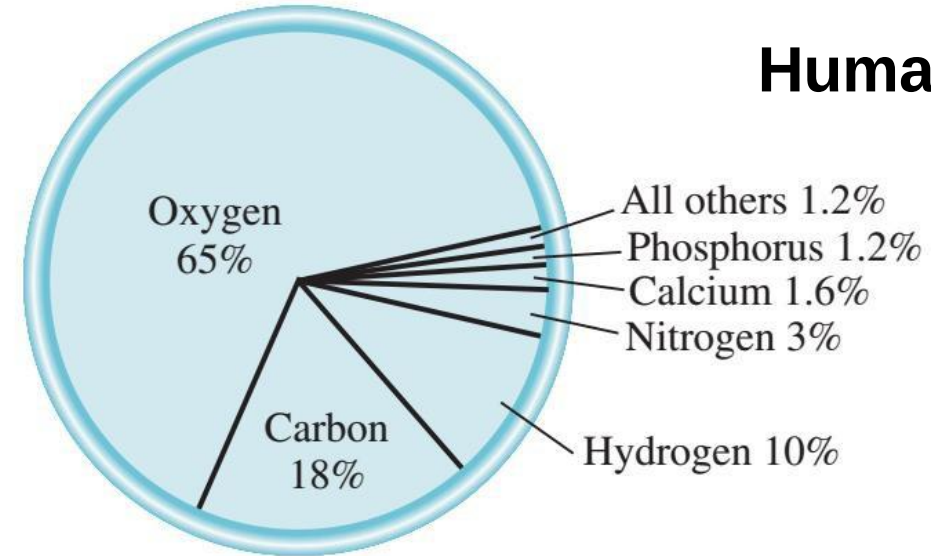
名	称：	7喜 冰爽柠檬味汽水
配	料：	水、果葡糖浆、食品添加剂（二氧化碳、柠檬酸、食用香精、柠檬酸钠、苯甲酸钠、安赛蜜、蔗糖素）。
净含量/规格：		7.92升（330毫升×24）

Distribution of Elements

Earth crust



Human body



• 纪念珍藏版 •

乡村教师

刘慈欣科幻自选集

刘慈欣◎著

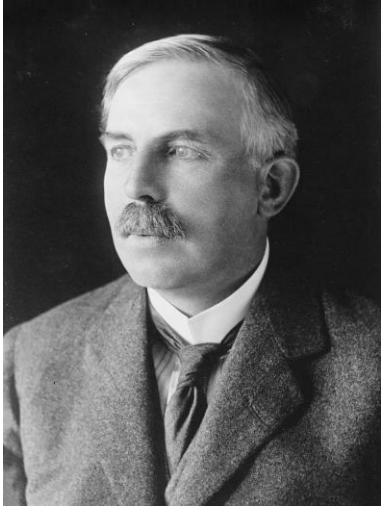
- “数百光年外，一场持续两万多年，涉及整个银河的战争即将结束。获胜的**碳基联邦**为了防止敌人**硅基帝国**死灰复燃，决定以制造横跨数百光年的恒星空白地带来囚禁对手。这也代表着归属这些恒星的行星以及行星上的生命的灭亡。为了银河系中其他大多数的碳基生命，联邦的除星行动冷酷而无慈悲。唯一能逃过这场绝对毁灭的方法就是被发现的生命是否已经具备足够的文明水平并加以证明。”
- “临终前，李宝库拼尽最后一丝力量将**力学三大定律**定理篆刻进四个孩子的脑海中。他并不知道，自己的四位学生马上将被选为碳基联邦**文明等级测试**的询问样本。他更不知道，自己刚刚让学生们背下来的力学定律，最终从碳基联邦的除星行动中拯救了地球文明，以及整个太阳系。”

Major points of today's lecture

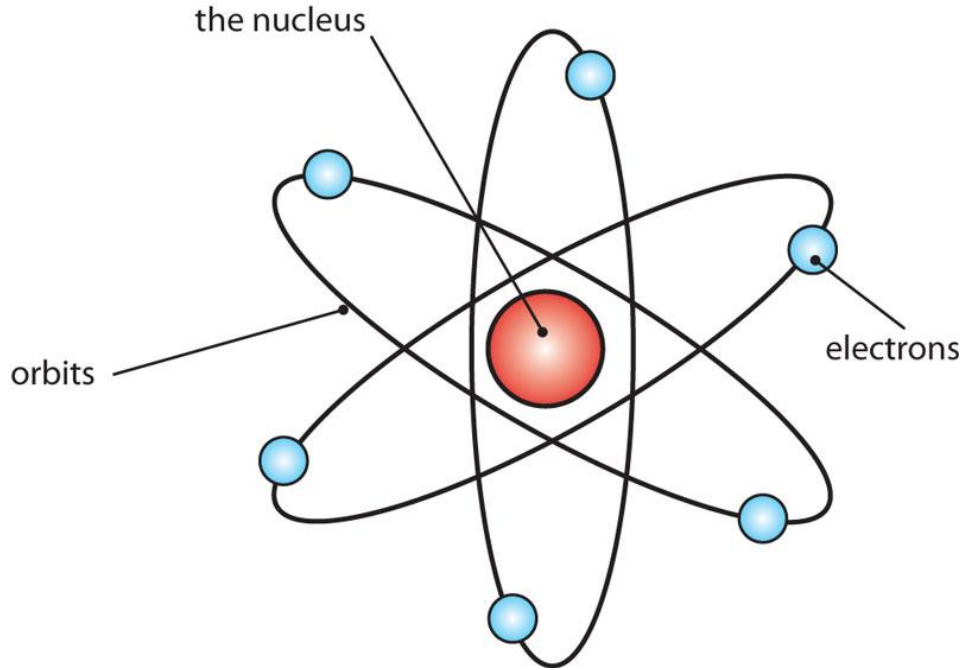
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Forces and potential energy in atoms



Ernest Rutherford
(1871-1937)



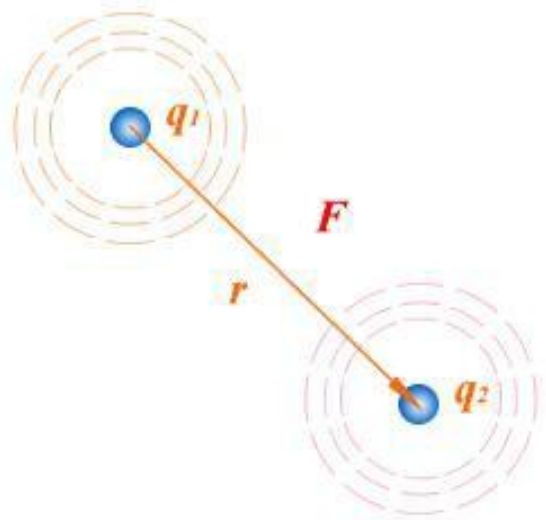
Rutherford's planetary model of the atom assumes that an atom of atomic number Z comprises a dense, central nucleus of **positive charge $+Ze$** surrounded **by a total of Z electrons** moving around the nucleus.

Structure and Behavior:

consequences of the **electrical forces** between the **electrons and the nucleus**.

Forces and potential energy in atoms

Coulomb's law(库仑定律)



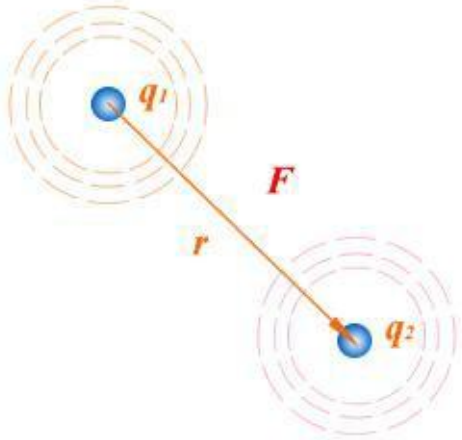
$$F(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$

ϵ_0 (vacuum permittivity, 真空介电常数), is a proportionality constant with a numerical value of $8.854 \cdot 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$.

Forces and potential energy in atoms

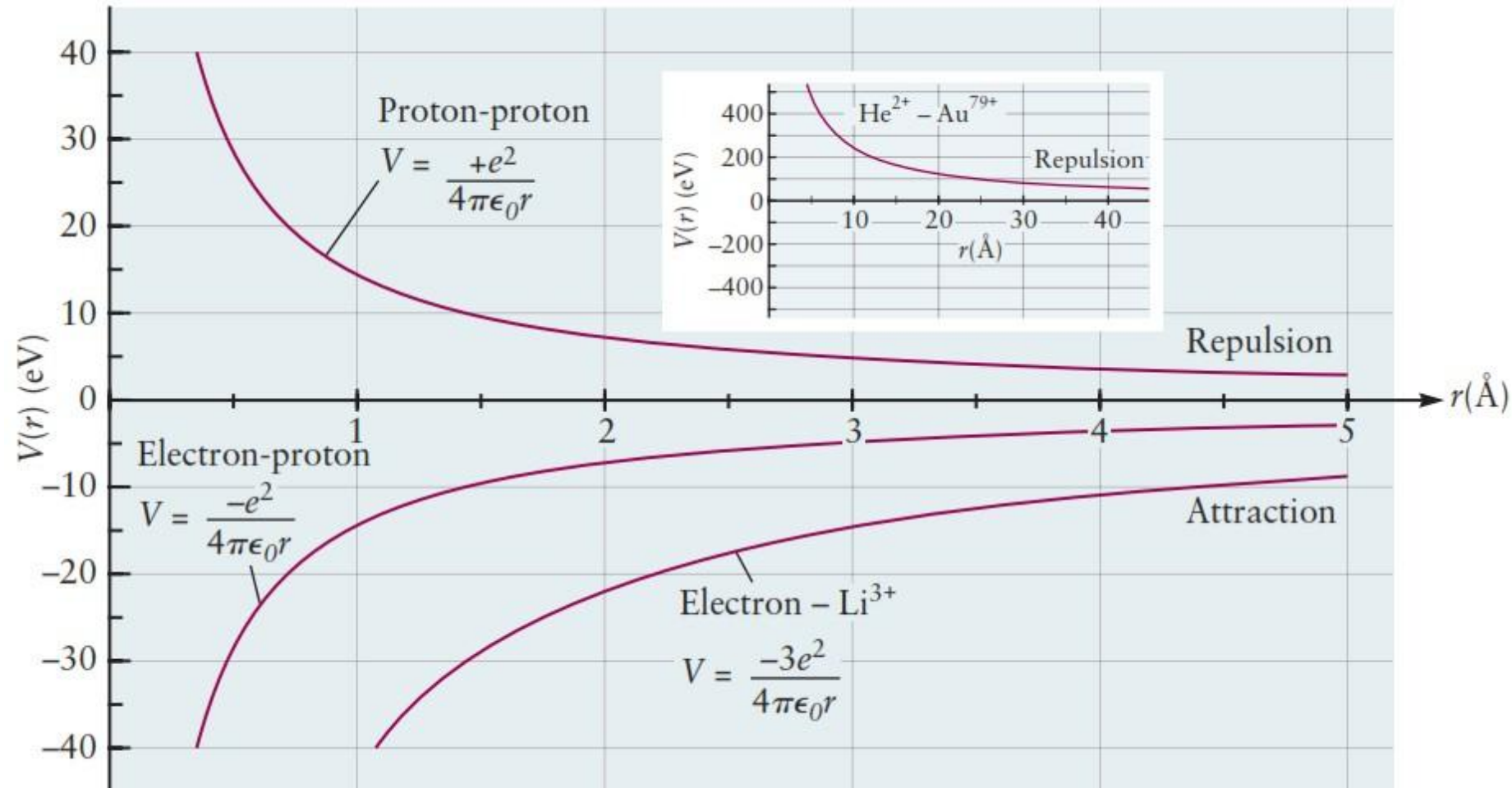
By convention, $V(r) \rightarrow 0$ as $r \rightarrow \infty$

$$F(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$$



$$V(r) = \frac{q_1 q_2}{4\pi\epsilon_0 r}$$

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$$



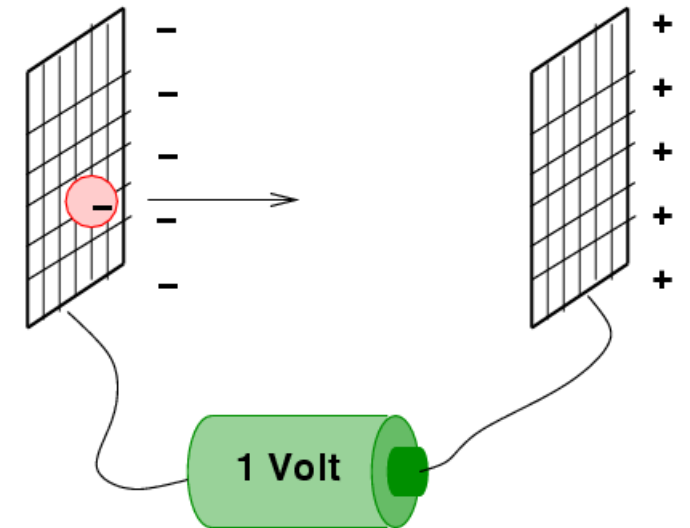
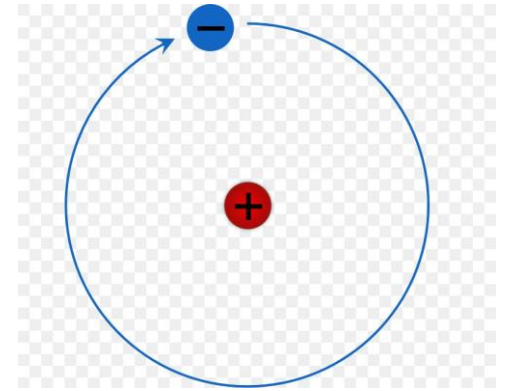
Forces and potential energy in atoms

- The typical **proton-electron distance** in a hydrogen atom is about 10^{-10} m.
- The **potential energy** of the hydrogen atom when the proton and electron are separated by a distance of $1 \text{ \AA}$ is

$$V(1\text{\AA}) = - \frac{(1.602 \times 10^{-19} \text{ C})^2}{4\pi(8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1})(1 \times 10^{-10} \text{ m})}$$
$$= - \frac{(8.988 \times 10^9)(1.602 \times 10^{-19})^2}{(1 \times 10^{-10})} \text{ J}$$

$$V(1\text{\AA}) = -2.307 \times 10^{-18} \text{ J}$$

$$V(1\text{\AA}) = - \frac{2.307 \times 10^{-18} \text{ J}}{1.602 \times 10^{-19} \text{ J(eV)}^{-1}} = -14.40 \text{ eV}$$

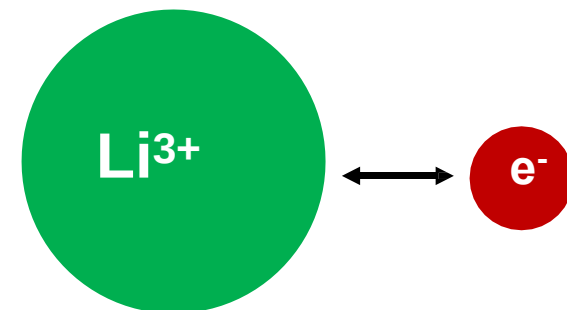


Electron volt 电子伏特

Question

Some sort of clamp is applied to hold an electron fixed at the distance of $2.5 \text{ \AA}$ from a lithium nucleus Li^{3+} which is also clamped in its position.

(a) Calculate the potential energy of the system of these two fixed particles.



Suppose in Figure 3.5 some sort of clamp is applied to hold an electron fixed at the distance of $2.5 \text{ \AA}$ from a lithium nucleus Li^{3+} which is also clamped in its position. (a) Calculate the potential energy of the system of these two fixed particles. (b) Calculate the magnitude of the force between the two fixed particles and state whether it is attractive or repulsive. (c) Predict the direction in which the particles will move if the clamps are suddenly removed.

Solution

(a) We use Equation 3.2 to calculate the potential energy as

$$\begin{aligned} V(2.5\text{\AA}) &= \frac{(+3e)(-e)(1.602 \times 10^{-19}\text{C})^2}{4\pi(8.854 \times 10^{-12}\text{C}^2 \text{J}^{-1} \text{m}^{-1})(2.5 \times 10^{-10} \text{m})} \\ &= -\frac{3(8.988 \times 10^9)(1.602 \times 10^{-19})^2}{(2.5 \times 10^{-10})} \text{J} \\ V(2.5\text{\AA}) &= -2.768 \times 10^{-18} \text{J} = -17.28 \text{eV} \end{aligned}$$

The potential energy at $2.5 \text{ \AA}$ is negative (i.e, lower than the value at infinite separation) because these oppositely charged particles attract one another.

(b) We calculate the force using the Coulomb law expression in Equation 3.1.

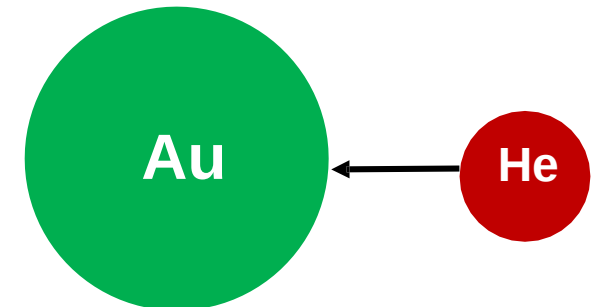
$$\begin{aligned} F(2.5\text{\AA}) &= \frac{(+3e)(-e)(1.602 \times 10^{-19}\text{C})^2}{4\pi(8.854 \times 10^{-12}\text{C}^2 \text{J}^{-1} \text{m}^{-1})(2.5 \times 10^{-10} \text{m})^2} \\ &= -\frac{3(8.988 \times 10^9)(1.602 \times 10^{-19})^2}{(2.5 \times 10^{-10})^2} \text{N} \\ F(2.5\text{\AA}) &= -2.304 \times 10^{-9} \text{N} \end{aligned}$$

The SI unit of force is the **newton** (N); $1 \text{ N} = 1 \text{ kg m s}^{-2}$. The negative sign means that the force is directed inward, towards the proton, so it is attractive.

(c) In Figure 3.5 the slope of the potential energy curve for these two particles is everywhere positive, including the position $r = 2.5 \text{ \AA}$. The force is directed inward (to the left) in regions where the slope is positive. The electron will move toward the Li^{3+} nucleus when the clamps holding the particles are released.

Calculation

Calculate the turning point of the alpha particles ($Z = 2$) arriving with initial kinetic energy of 2.0 MeV when they collided with gold nuclei ($Z = 79$) in Rutherford's experiment. Estimate the size of the gold nuclei.



Calculate the turning point of the alpha particles ($Z = 2$) arriving with initial kinetic energy of 2.0 MeV when they collided with gold nuclei ($Z = 79$) in Rutherford's experiment. Estimate the size of the gold nuclei.

Solution

At the turning point r_{tp} the potential energy is equal to the initial kinetic energy of the arriving alpha particle. From Equation 3.2 we obtain the relation

$$r_{\text{tp}} = \frac{q_1 q_2}{4\pi\epsilon_0 E}$$

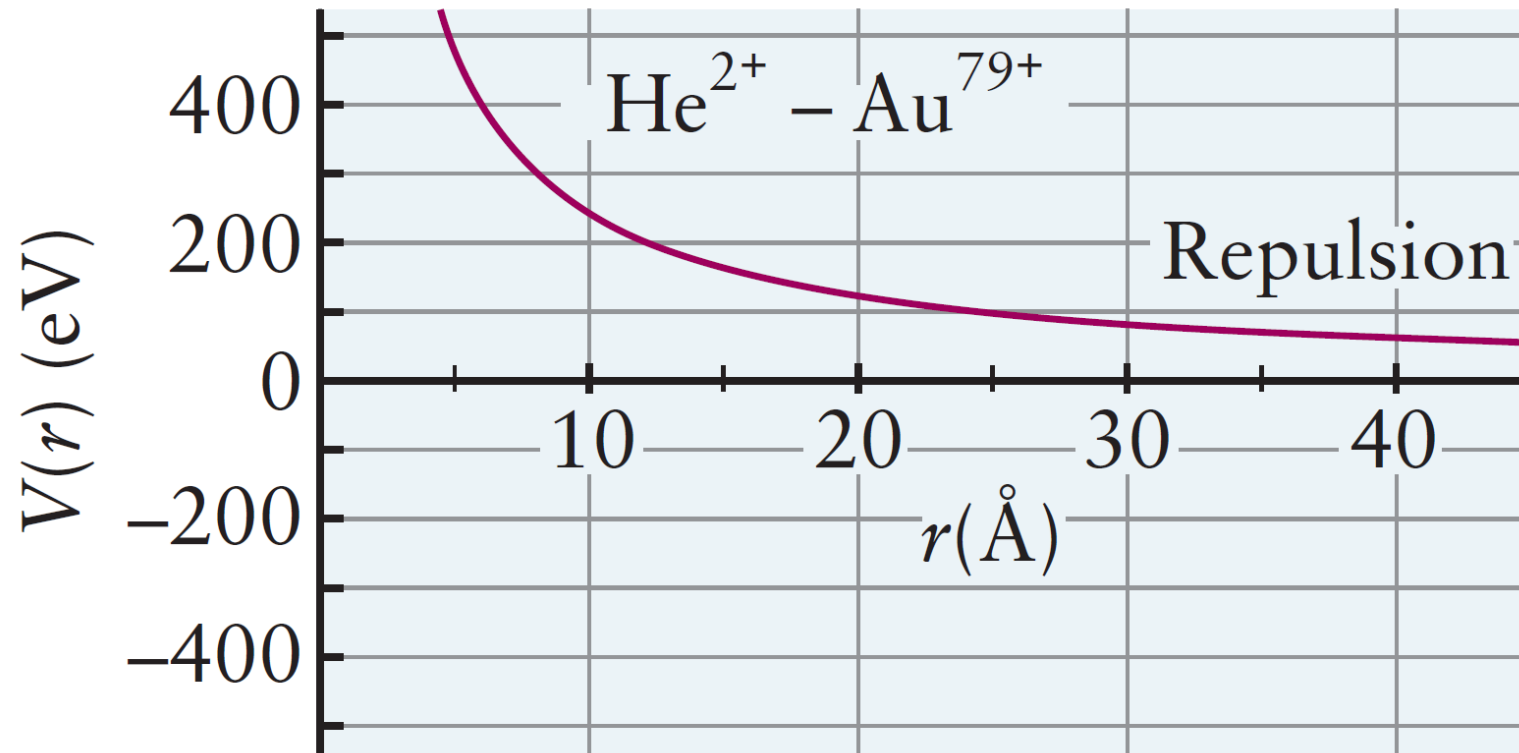
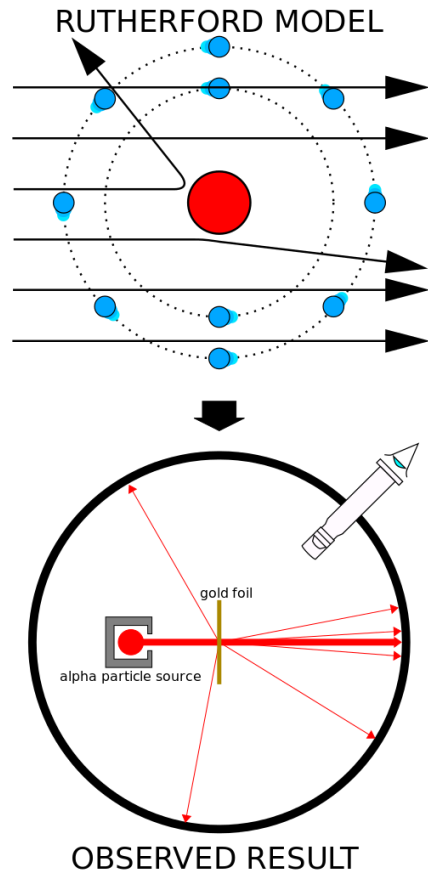
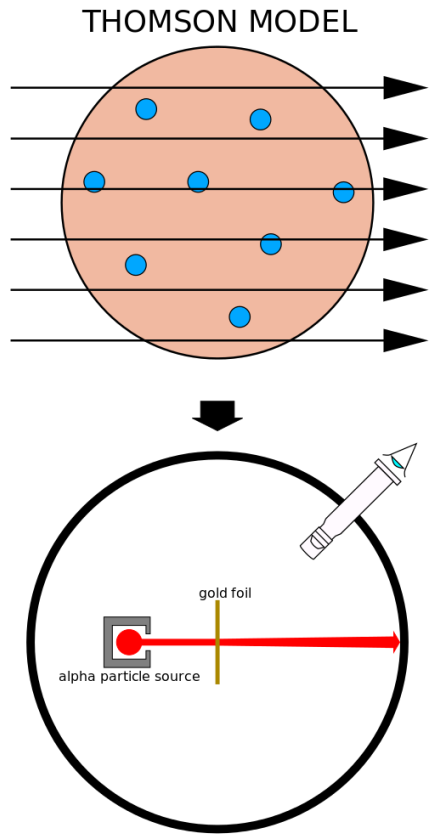
$$r_{\text{tp}} = \frac{(+2e)(+79e)(1.602 \times 10^{-19} \text{C})^2}{4\pi(8.854 \times 10^{-12} \text{C}^2 \text{J}^{-1} \text{m}^{-1})(2.0 \times 10^6 \text{eV})}$$

$$r_{\text{tp}} = \frac{(2)(79)(8.988 \times 10^9)(1.602 \times 10^{-19})^2 \text{J m}}{(2.0 \times 10^6 \text{eV})(1.602 \times 10^{-19} \text{J eV}^{-1})}$$

$$r_{\text{tp}} = 11.4 \times 10^{-14} \text{ m} = 11.4 \times 10^{-12} \text{ cm} = 11.4 \times 10^{-4} \text{ \AA}$$

At the turning point we assume the alpha particle is touching the gold nucleus but not penetrating into it. The radius of the gold nucleus then must be less than $11.4 \times 10^{-14} \text{ m}$. Other experimental and theoretical studies confirm that 10^{-14} m is a good estimate for the size of atomic nuclei.

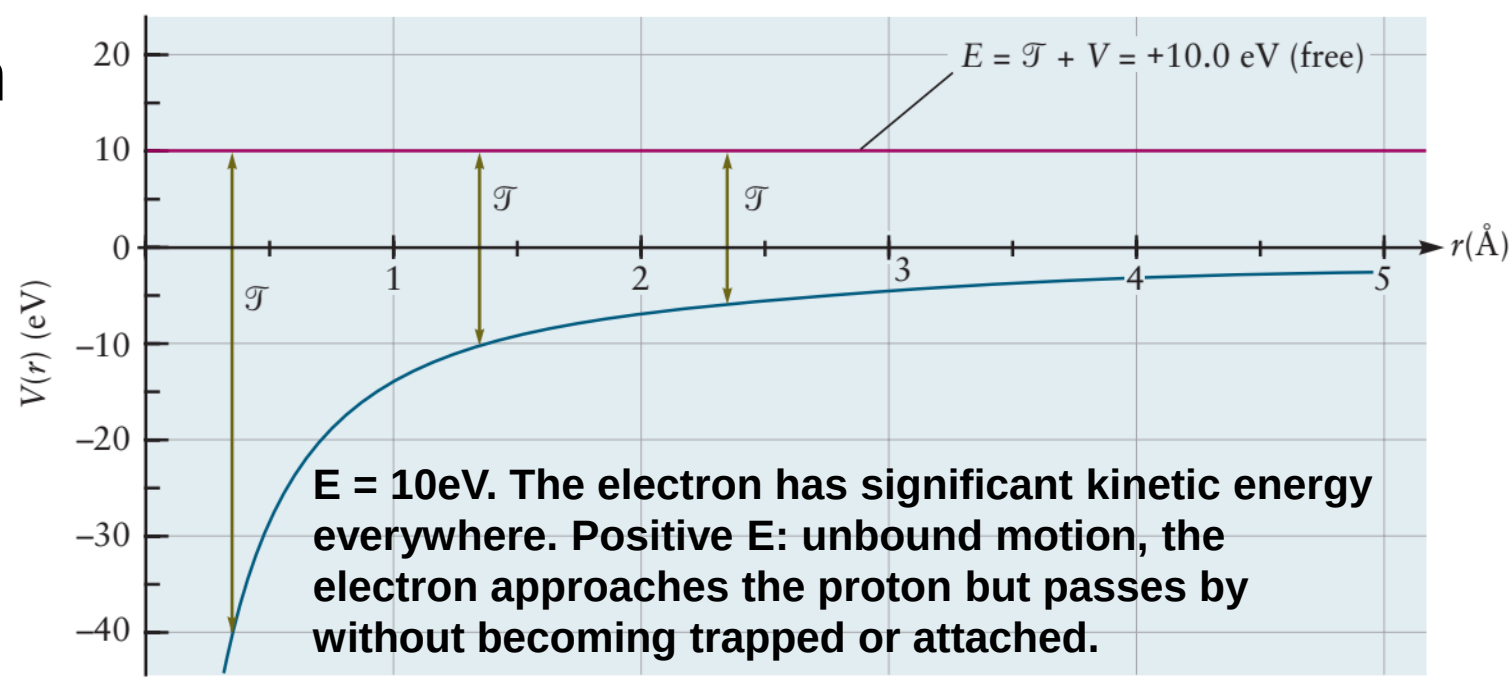
Kinetic energy transfer



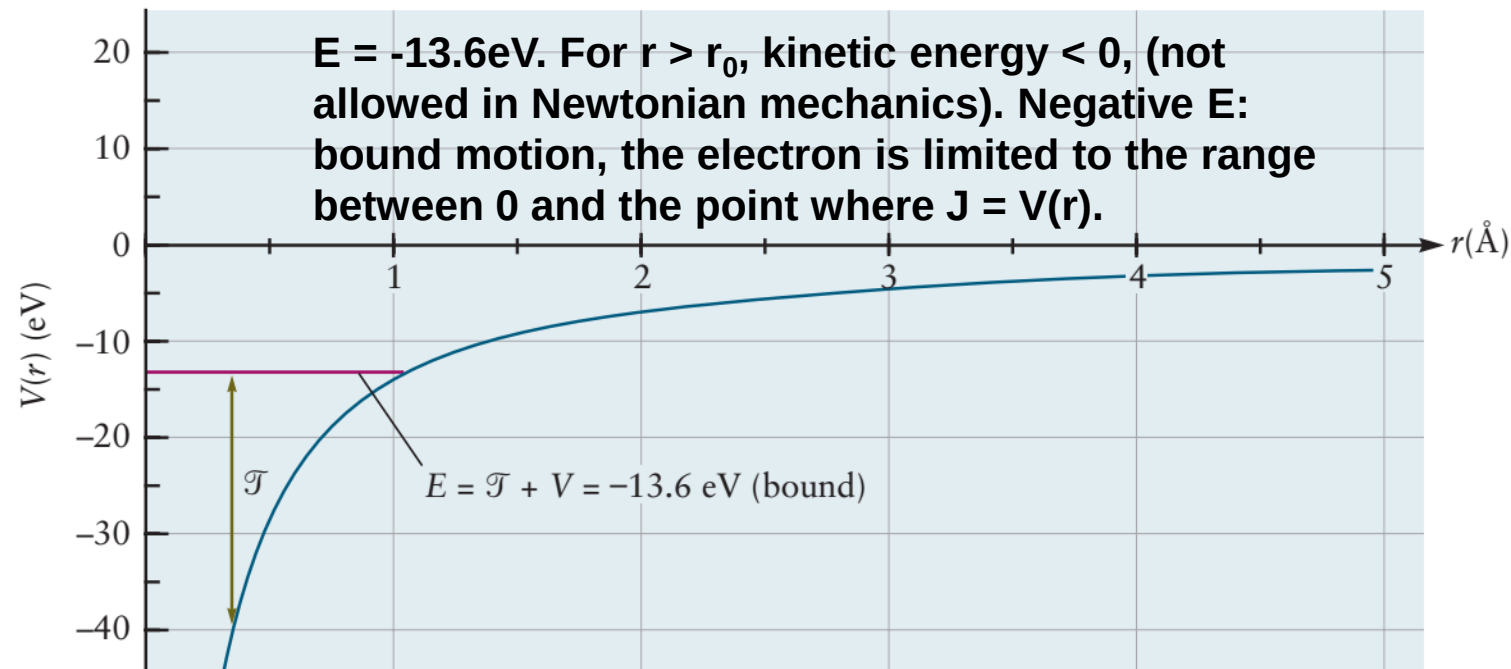
Total energy of electron in Hydrogen atom

The total energy (**kinetic + potential**) of the electron in the atom is

$$E = \frac{1}{2} m_e v^2 - \frac{Z e^2}{4 \pi \epsilon_0 r}$$



(a)



Calculation

Assume a hydrogen atom has bound state energy $E = -13.6 \text{ eV}$.

Calculate the maximum distance r_{max} of the electron from the nucleus allowed by Newtonian mechanics.



Assume a hydrogen atom has bound state energy $E = -13.6$ eV. Calculate the maximum distance $r_{\max}$ of the electron from the nucleus allowed by Newtonian mechanics.

Solution

When the electron is located at $r_{\max}$ the potential energy is equal to the bound state energy of the atom. From Equation 3.3 we obtain the relations

$$V(r_{\max}) = -\frac{e^2}{4\pi\epsilon_0 r_{\max}} = -13.6 \text{ eV}$$

$$r_{\max} = \frac{e^2}{4\pi\epsilon_0 (13.6 \text{ eV})}$$

$$r_{\max} = \frac{(1.602 \times 10^{-19} \text{ C})^2}{4\pi(8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1})(13.6 \text{ eV})(1.602 \times 10^{-19} \text{ J eV}^{-1})}$$

$$r_{\max} = 1.06 \times 10^{-10} \text{ m} = 1.06 \text{ \AA}$$

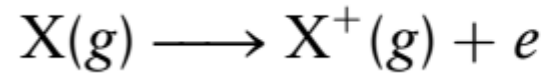
It is tempting to interpret $r_{\max}$ as a measure of the “size” of the hydrogen atom, but that idea has to be refined when we describe the hydrogen atom by quantum mechanics. In Chapter 5 we see that quantum mechanics describes the position of the electron around the nucleus by a probability function and that 99% of the probability lies within a sphere of radius $2.2\text{\AA}$ centered on the nucleus when the bound state energy is -13.6 eV. Quantum mechanics allows small particles such as electrons to penetrate into regions where classical mechanics forbids them to go.

Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Representations of molecules
 2. The periodic table
 3. Forces and potential energy in atoms
 4. Ionization energies, the shell model of the atom, and shielding
 5. Electron affinity
 6. Electronegativity: the tendency of atoms to attract electrons in molecules

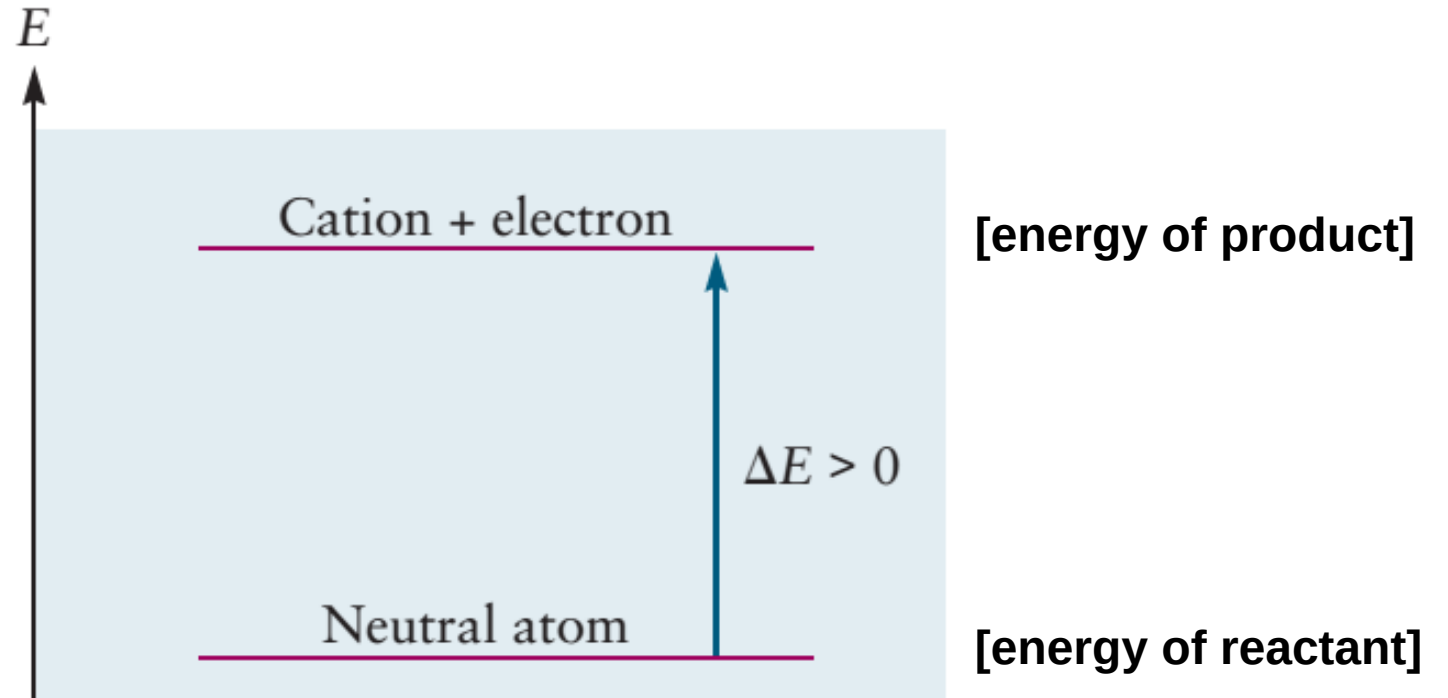
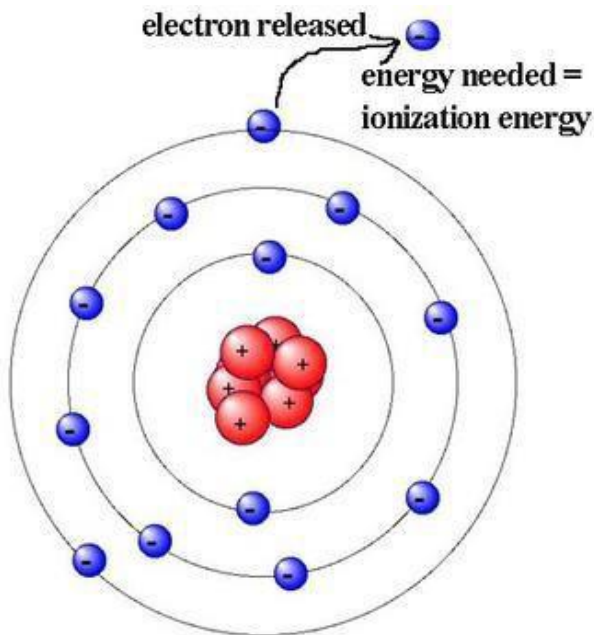
Ionization Energies (电离能)

The **ionization energy**, IE_1 , of an atom (first ionization energy) is the minimum energy necessary to detach an electron from the neutral gaseous atom and form a positively charged gaseous ion.



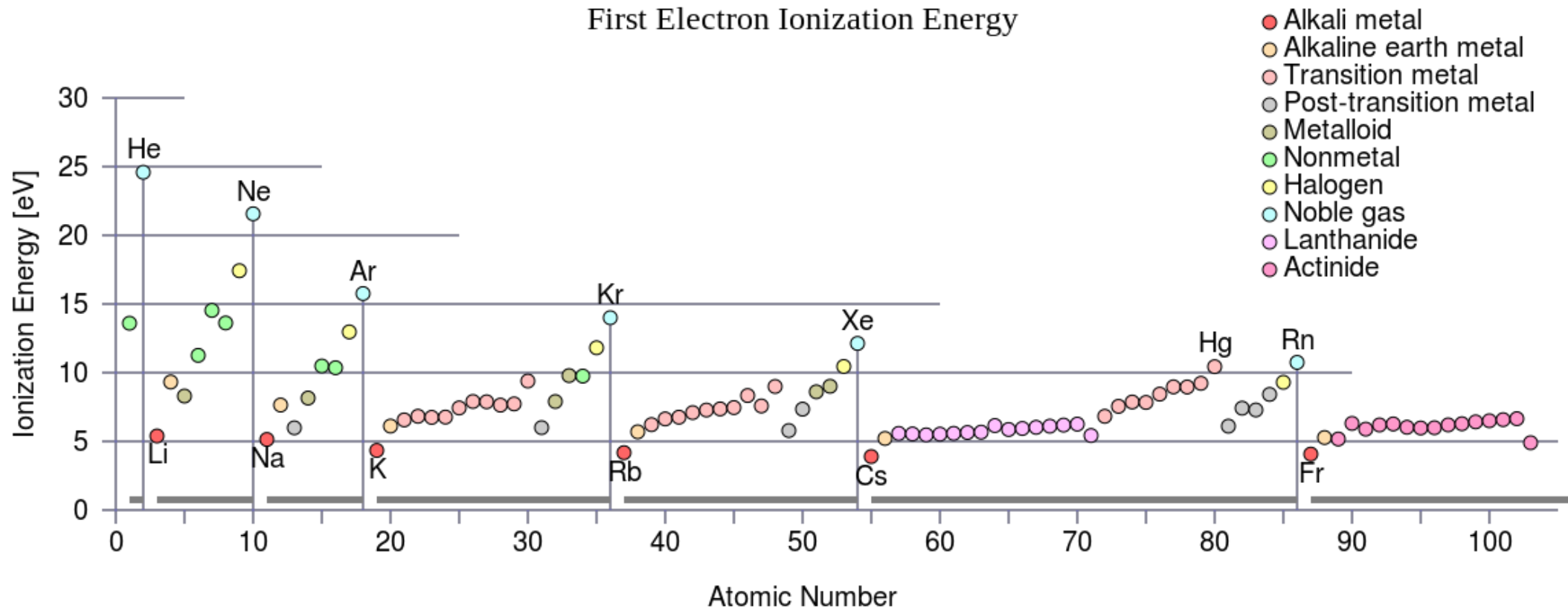
$$\Delta E = IE_1$$

$$\Delta E = [\text{energy of reaction products}] - [\text{energy of reactants}]$$



Ionization Energies (IE)

- ΔE for ionization reactions is always **positive**.
- The ionization energy is a measure of the **stability** of the **free atom**.
- Atoms with **larger ionization** energies are **more stable**.



Ionization Energies (IE)

The **second ionization energy**, IE_2 : is the minimum energy required to remove a second electron.



- **The third, fourth, and higher ionization energies** are defined in analogous fashion.
- Successive ionization energies always **increase** due to the **greater electrostatic attraction** of the electron to the product ions, which have increasingly greater positive charges.

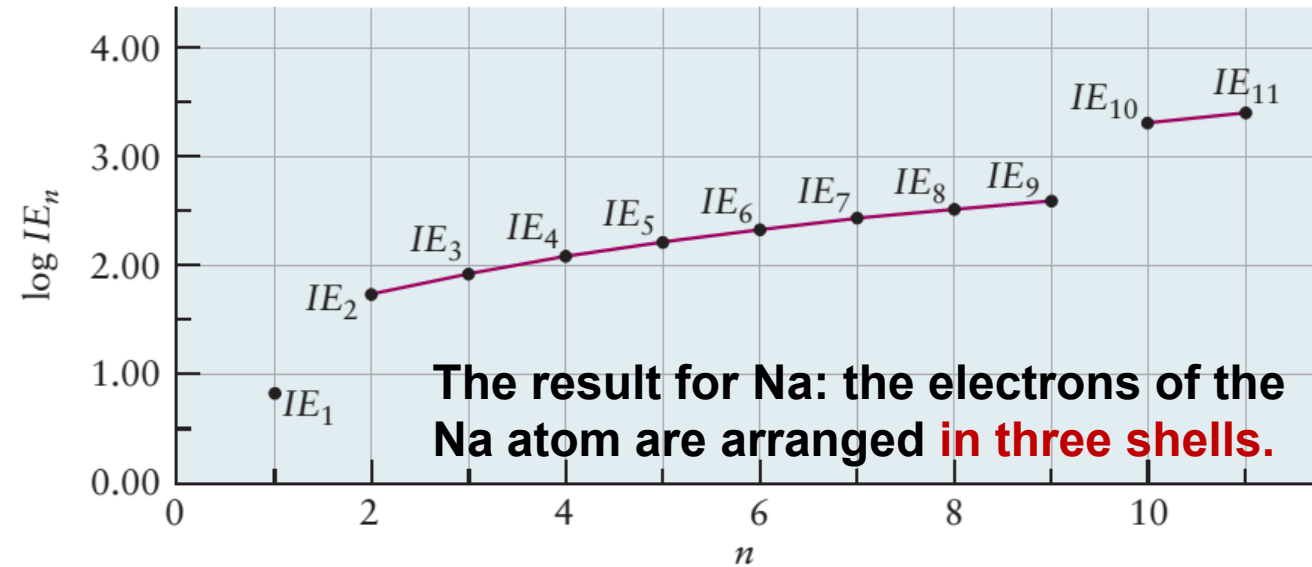
Ionization Energies (IE)

Z	Element	IE_1	IE_2	IE_3	IE_4	IE_5	IE_6	IE_7	IE_8	IE_9	IE_{10}
1	H	13.60									
2	He	24.59	54.42								
3	Li	5.39	75.64	122.45							
4	Be	9.32	18.21	153.89	217.71						
5	B	8.30	25.15	37.93	259.37	340.22					
6	C	11.26	24.38	47.89	64.49	392.08	489.99				
7	N	14.53	29.60	47.45	77.47	97.89	552.06	667.03			
8	O	13.62	35.12	54.93	77.41	113.90	138.12	739.32	871.39		
9	F	17.42	34.97	62.71	87.14	114.24	157.16	185.18	953.89	1103.08	
10	Ne	21.56	40.96	63.45	97.11	126.21	157.93	207.27	239.09	1195.79	1362.16
11	Na	5.14	47.29	71.64	98.91	138.39	172.15	208.47	264.18	299.87	1465.10
12	Mg	7.65	15.04	80.14	109.24	141.26	186.50	224.94	265.90	327.94	367.53
13	Al	5.99	18.83	28.45	119.99	153.71	190.47	241.43	284.59	330.21	398.57
14	Si	8.15	16.35	33.49	45.14	166.77	205.05	246.52	303.17	351.10	401.43
15	P	10.49	19.73	30.18	51.37	65.02	220.43	263.22	309.41	371.73	424.50
16	S	10.36	23.33	34.83	47.30	72.68	88.05	280.93	328.23	379.10	447.10
17	Cl	12.97	23.81	39.61	53.46	67.8	97.03	114.19	348.28	400.03	455.62
18	Ar	15.76	27.63	40.74	59.81	75.02	91.01	124.32	143.46	422.43	478.68
19	K	4.34	31.63	45.72	60.91	82.66	100.0	117.56	154.86	175.82	503.44
20	Ca	6.11	11.87	50.91	67.10	84.41	108.78	127.7	147.24	188.54	211.27
21	Sc	6.54	12.80	24.76	73.47	91.66	111.1	138.0	158.7	180.02	225.32

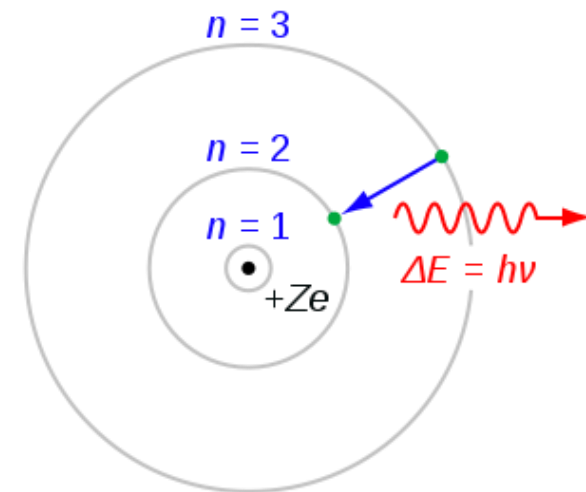
Electrons in the atoms of (Li, Be, B, C, N, O, F, Ne) are arranged as a stable **helium-like inner core**, surrounded by less tightly bound electrons. (**Period 2**)

Ionization Energies (IE)

Z	Element	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	IE ₇	IE ₈	IE ₉	IE ₁₀
1	H	13.60									
2	He	24.59	54.42								
3	Li	5.39	75.64	122.45							
4	Be	9.32	18.21	153.89	217.71						
5	B	8.30	25.15	37.93	259.37	340.22					
6	C	11.26	24.38	47.89	64.49	392.08	489.99				
7	N	14.53	29.60	47.45	77.47	97.89	552.06	667.03			
8	O	13.62	35.12	54.93	77.41	113.90	138.12	739.32	871.39		
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11	Na	5.14	47.29	71.64	98.91	138.39	172.15	208.47	264.18	299.87	1465.10
12	Mg	7.65	15.04	80.14	109.24	141.26	186.50	224.94	265.90	327.94	367.53
13	Al	5.99	18.83	28.45	119.99	153.71	190.47	241.43	284.59	330.21	398.57
14	Si	8.15	16.35	33.49	45.14	166.77	205.05	246.52	303.17	351.10	401.43
15	P	10.49	19.73	30.18	51.37	65.02	220.43	263.22	309.41	371.73	424.50
16	S	10.36	23.33	34.83	47.30	72.68	88.05	280.93	328.23	379.10	447.10
17	Cl	12.97	23.81	39.61	53.46	67.8	97.03	114.19	348.28	400.03	455.62
18	Ar	15.76	27.63	40.74	59.81	75.02	91.01	124.32	143.46	422.43	478.68
19	K	4.34	31.63	45.72	60.91	82.66	100.0	117.56	154.86	175.82	503.44
20	Ca	6.11	11.87	50.91	67.10	84.41	108.78	127.7	147.24	188.54	211.27
21	Sc	6.54	12.80	24.76	73.47	91.66	111.1	138.0	158.7	180.02	225.32

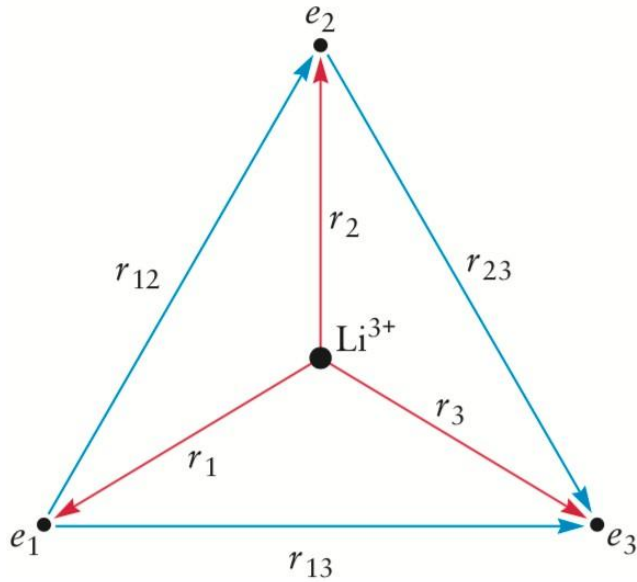


- The **first electron** is easily removed to produce Na⁺, which has **neon like stability** as indicated by the large jump between IE₁ and IE₂.
- **Electrons 2 through 9** occupy the **second shell**, and all of them are more tightly bound than the first electron.
- A big jump between IE₉ and IE₁₀ suggests that the last two electrons occupy a **third shell**, the electrons of which are the most tightly bound of all.



<i>Z</i>	Element	<i>IE</i> ₁	<i>IE</i> ₂	<i>IE</i> ₃	<i>IE</i> ₄	<i>IE</i> ₅	<i>IE</i> ₆	<i>IE</i> ₇	<i>IE</i> ₈	<i>IE</i> ₉	<i>IE</i> ₁₀
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7	N	14.53	29.60	47.45	77.47	97.89	552.06	667.03			
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21	Sc	6.54	12.80	24.76	73.47	91.66	111.1	138.0	158.7	180.02	225.32

From one electron atom to multi-electron atom



Li atom with three electrons.

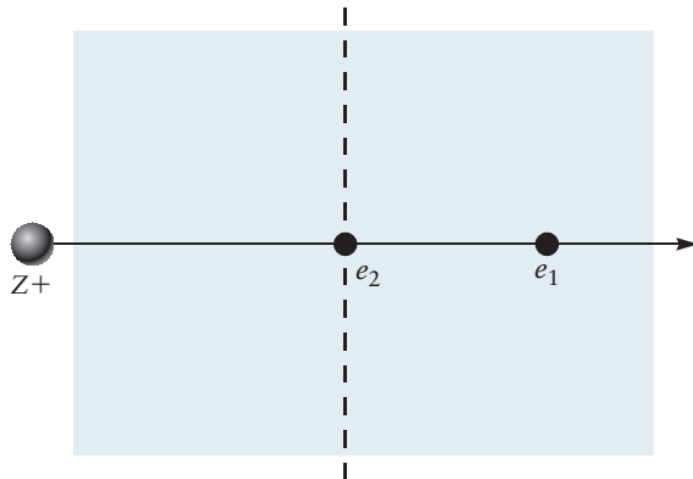
$$V = \frac{e^2}{4\pi\epsilon_0} \left(-\frac{Z}{r_1} - \frac{Z}{r_2} - \frac{Z}{r_3} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{23}} \right)$$

Potential energy

How to simplify?



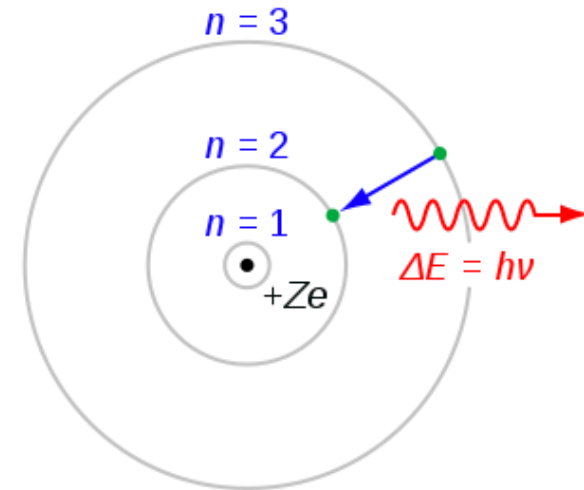
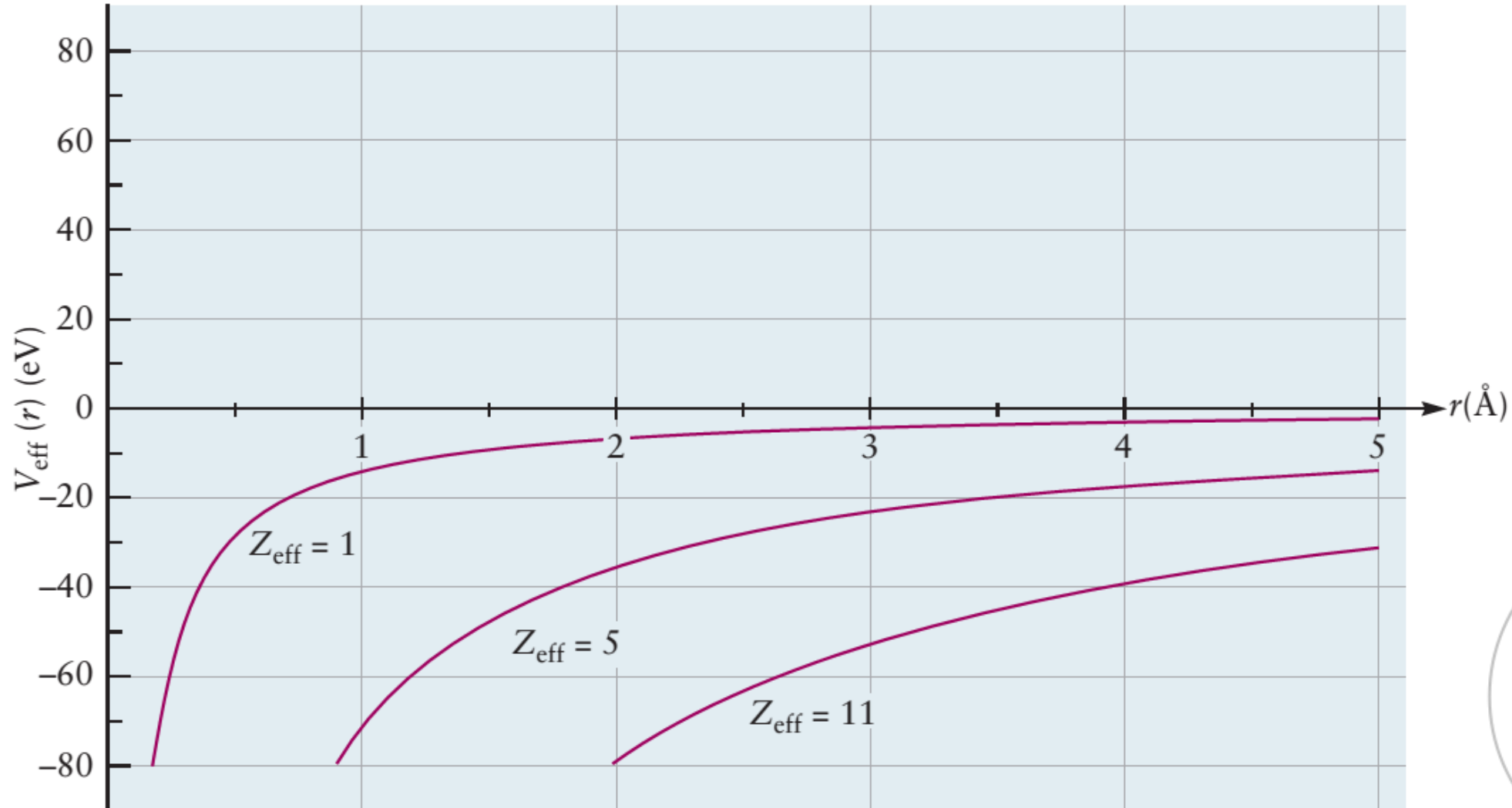
Effective potential energy = attractive nuclear-electron force + repulsive electron-electron force



$Z_{\text{eff}} = Z - S$ 有效原子序数

$$V_{\text{eff}}(r) = -\frac{Z_{\text{eff}}e^2}{4\pi\epsilon_0 r}$$

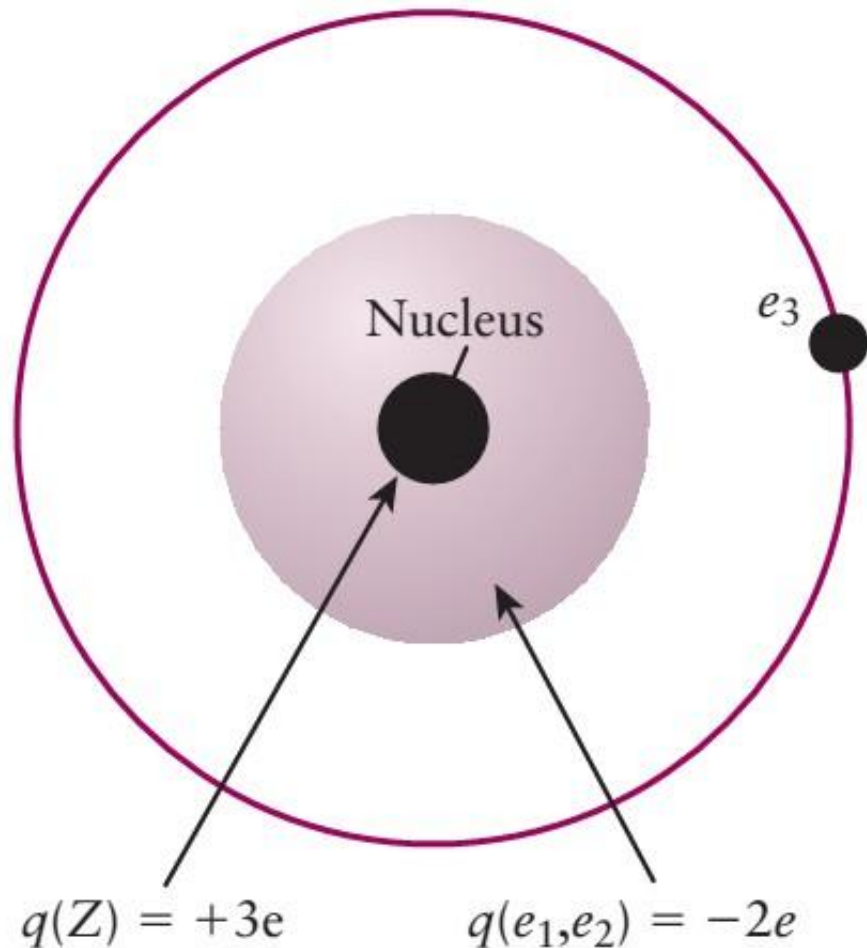
The shell Model of the atom



- **Core electrons** do not participate significantly in chemical reactions.
- **Valence electrons** involved in chemical bonding.

The shell Model of the atom

Lewis (路易斯) dot symbol



- The valence electrons as dots arranged around the chemical symbol
- The core electrons are not shown.
- The first four dots are arranged individually around the four sides of the symbol.
- If an atom has more than four valence electrons, dots representing them are then paired with those already present.

·H

·He

·Li

·Be·

·B·

·C·

·N·

·O·

·F·

·Ne·

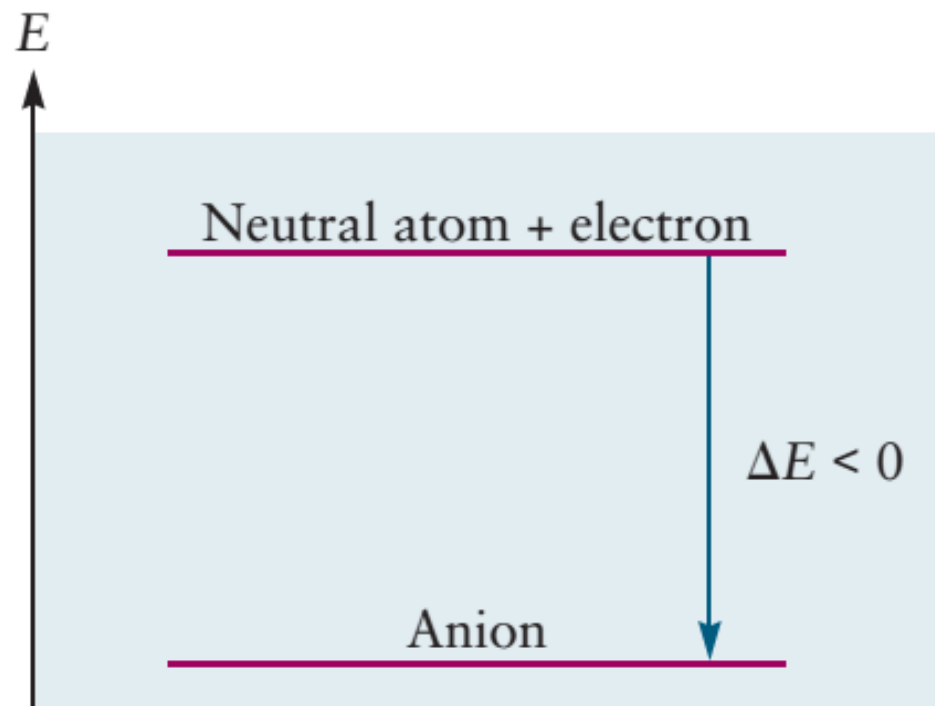
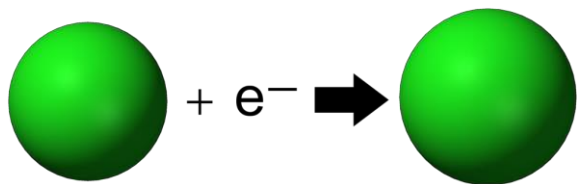
Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**

1. Representations of molecules
2. The periodic table
3. Forces and potential energy in atoms
4. Ionization energies, the shell model of the atom, and shielding
5. **Electron affinity**
6. **Electronegativity: the tendency of atoms to attract electrons in molecules**

Electron affinity

Electron attachment energy (电子附着能): an atom accepts an extra electron to form an anion.

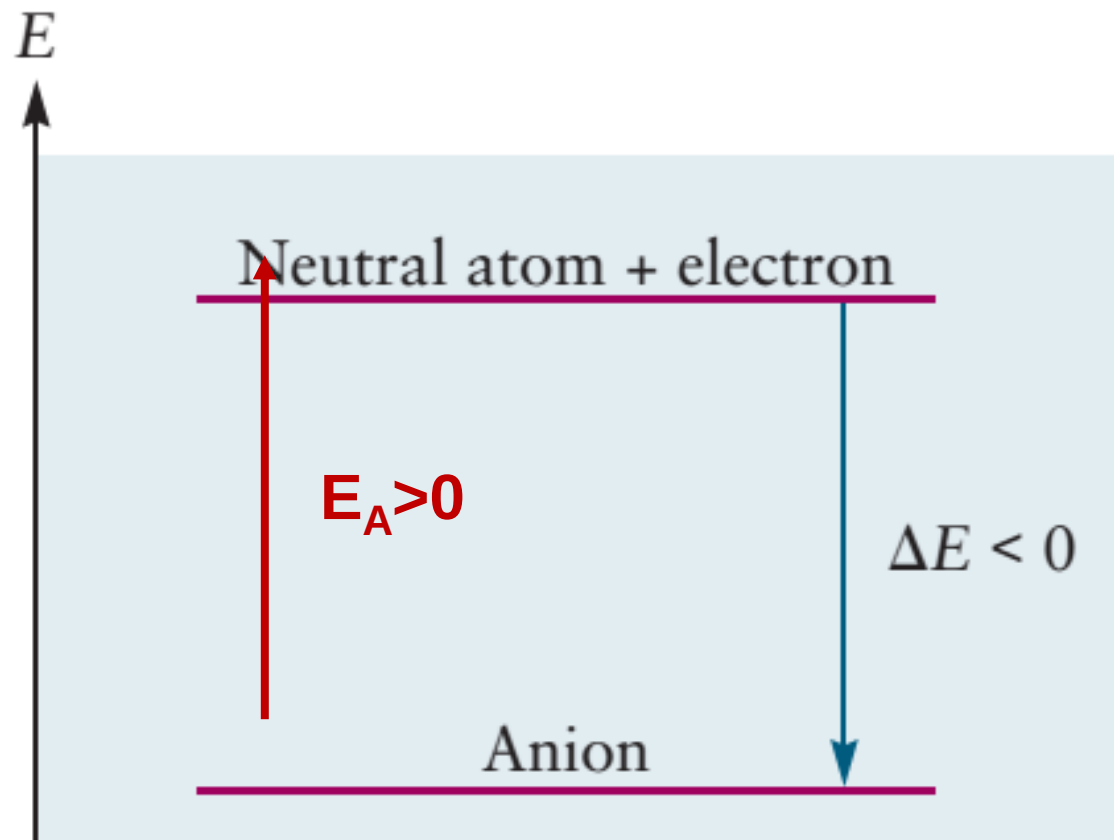
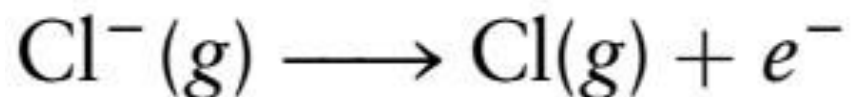


Usually, $\Delta E < 0$

Difficult to measure electron attachment energy!

Electron affinity

Electron affinity (EA_X) (电子亲合能) of atom X: energy required to detached the electron from the anion X^- and give the neutral atom.

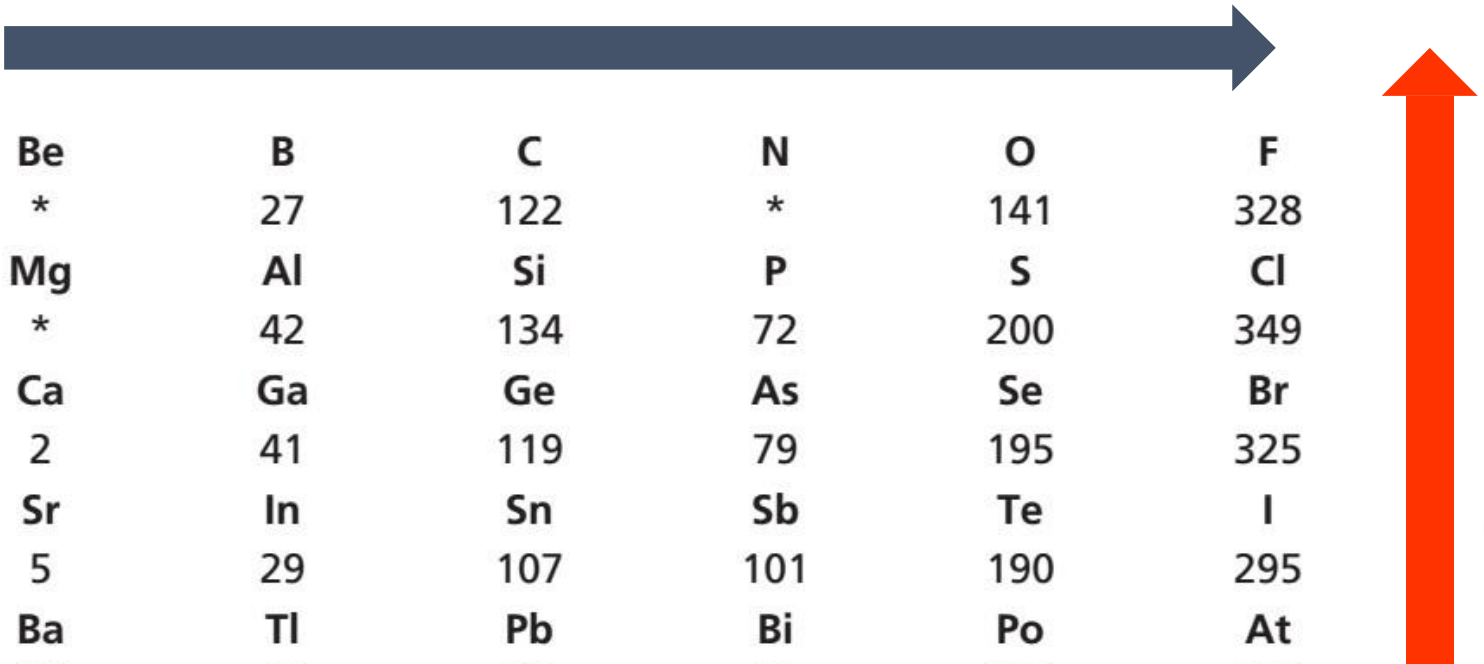


$$EA_{Cl} = 349 \text{ kJ/mol}$$

Electron affinity

TABLE 3.2

Electron Affinity of Selected Atoms (in kJ mol^{-1})



H								He
73								*
Li	Be	B	C	N	O	F		Ne
60	*	27	122	*	141	328		*
Na	Mg	Al	Si	P	S	Cl		Ar
53	*	42	134	72	200	349		*
K	Ca	Ga	Ge	As	Se	Br		Kr
48	2	41	119	79	195	325		*
Rb	Sr	In	Sn	Sb	Te	I		Xe
47	5	29	107	101	190	295		*
Cs	Ba	Tl	Pb	Bi	Po	At		Rn
46	14	19	35	91	183	270		*

*No stable anion A^- exists for this element in the gas phase.

No gaseous atom has a positive electron affinity for a second electron.

Electron affinity comparison

19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
---------	----------	----------	----------	---------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Higher Z_{eff}
Same distance



Higher electron
ionization energy

Higher Z_{eff}
Same distance



Ion is more
stable



Higher electron
affinity

Consider the elements selenium (Se) and bromine (Br). Without consulting tables of data, predict which one should have the higher value of ionization energy IE_1 . Predict which should have the higher value of electron affinity EA .

Solution

These elements are in adjacent groups in the 4th period of the periodic table. Progressing from selenium to bromine adds one unit of positive charge, which increases Z_{eff} . At the same time we add one more electron into the 4th shell like the valence electrons in Se. Since the nuclear attraction is greater for the added electron in bromine and its distance from the nucleus is about the same, it will be more tightly bound than the valence electrons in selenium. Therefore bromine will have the larger value of IE_1 .

If we place an additional electron in the 4th shell of each atom to form an anion, the additional electron on bromine will experience a larger value of Z_{eff} , and the Br^- anion will have greater stability than the Se^- anion. Moreover, Br^- has a stable octet configuration whereas Se^- does not. So Br has the larger value of EA .

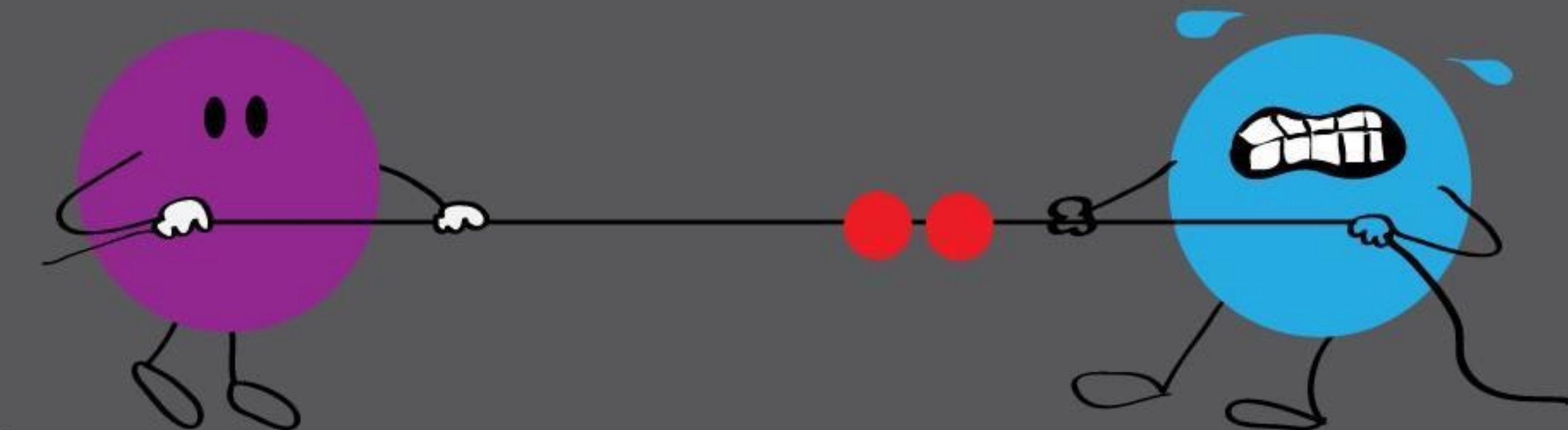
Major points of today's lecture

- **Atomic shells and classical models of chemical bonding**
 1. Representations of molecules
 2. The periodic table
 3. Forces and potential energy in atoms
 4. Ionization energies, the shell model of the atom, and shielding
 5. Electron affinity
 6. Electronegativity: the tendency of atoms to attract electrons in molecules

ATOM A
weak pull

BIG
electronegativity
difference

ATOM B
strong pull



Can we define it ?

Electronegativity

Electronegativity (电负性): the tendency of atoms to attract electrons in molecules.

Mulliken's Electronegativity

$$\text{EN (Mulliken)} \propto \frac{1}{2} (IE_1 + EA)$$



Robert C. Mulliken
(1896–1986)

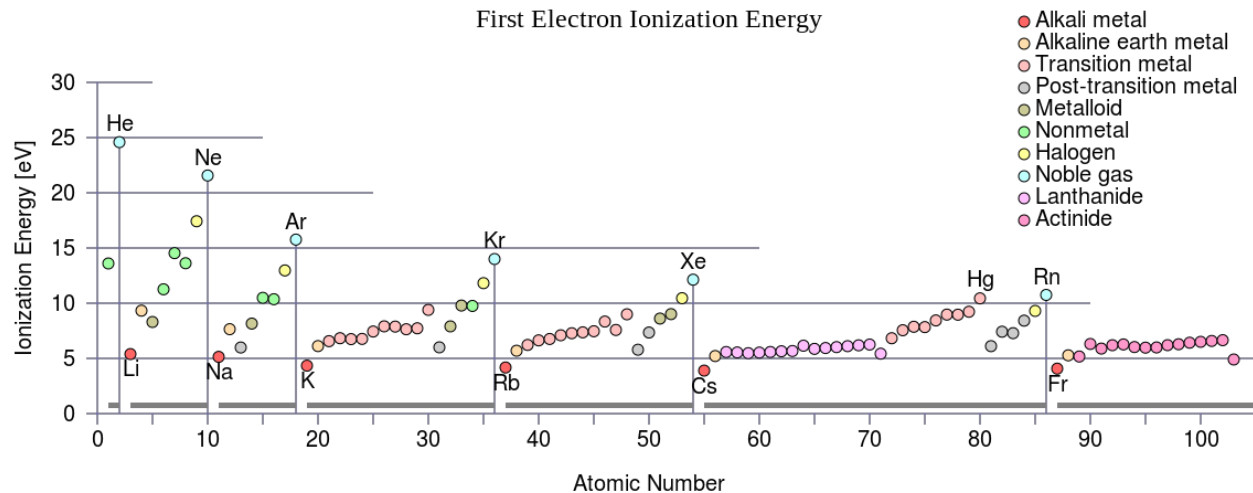
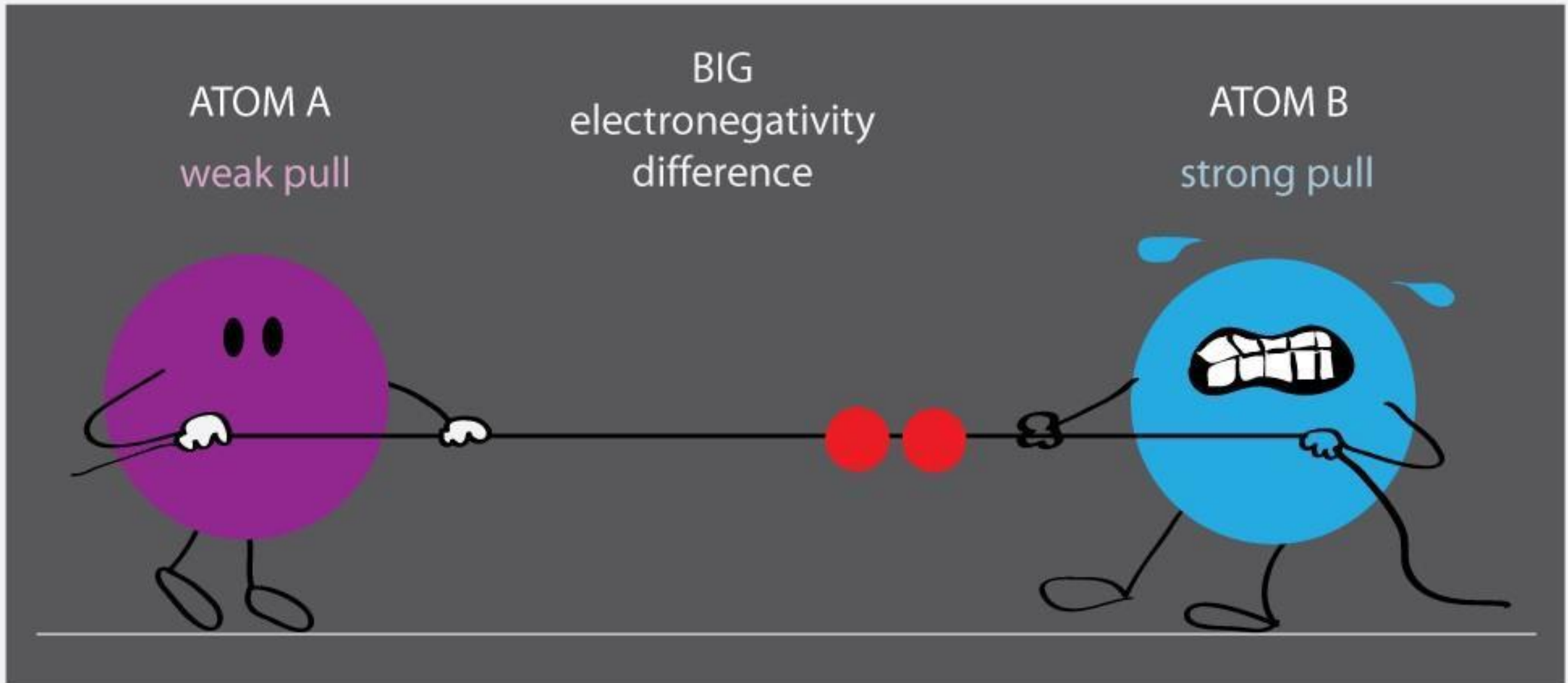


TABLE 3.2

Electron Affinity of Selected Atoms (in kJ mol⁻¹)

H							He
73							*
Li	Be	B	C	N	O	F	Ne
60	*	27	122	*	141	328	*
Na	Mg	Al	Si	P	S	Cl	Ar
53	*	42	134	72	200	349	*
K	Ca	Ga	Ge	As	Se	Br	Kr
48	2	41	119	79	195	325	*
Rb	Sr	In	Sn	Sb	Te	I	Xe
47	5	29	107	101	190	295	*
Cs	Ba	Tl	Pb	Bi	Po	At	Rn
46	14	19	35	91	183	270	*

*No stable anion A⁻ exists for this element in the gas phase.



- **Electron donors** have small ionization energies and small electron affinities (**low electronegativities**); they are highly **electropositive**.
- **Electron acceptors** have large ionization energies and large electron affinities (**high electronegativities**); they are highly **electronegative**.

The Nobel Prize in Chemistry 1966

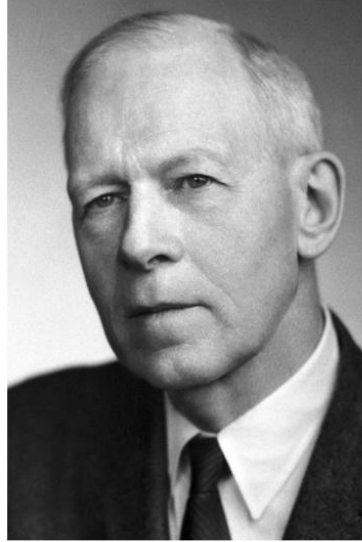
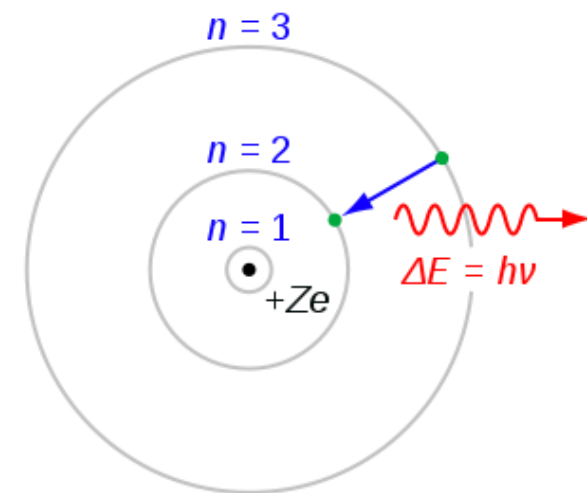
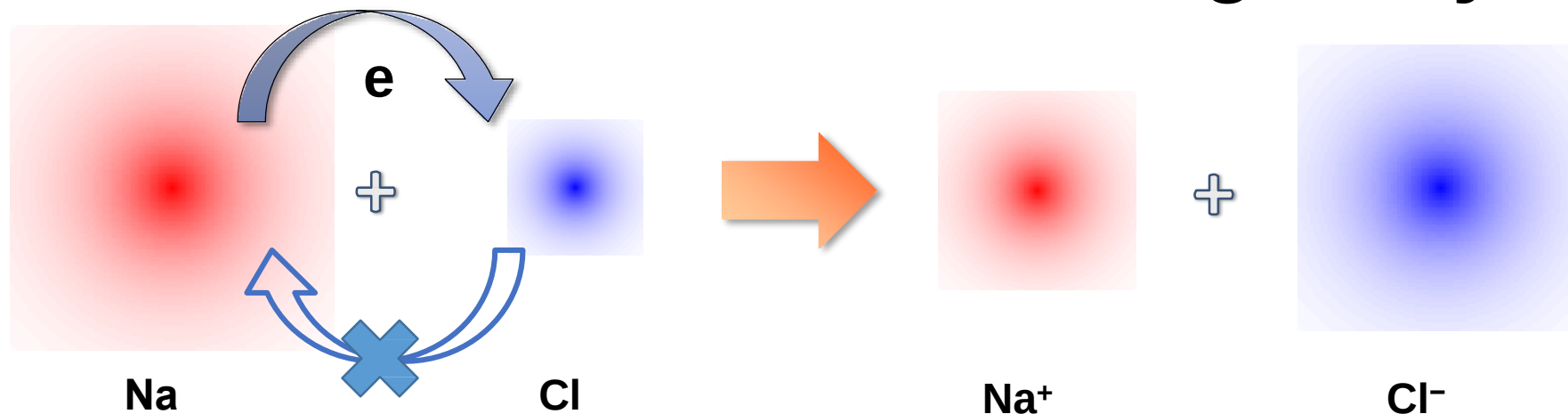


Photo from the Nobel
Foundation archive.
Robert S. Mulliken

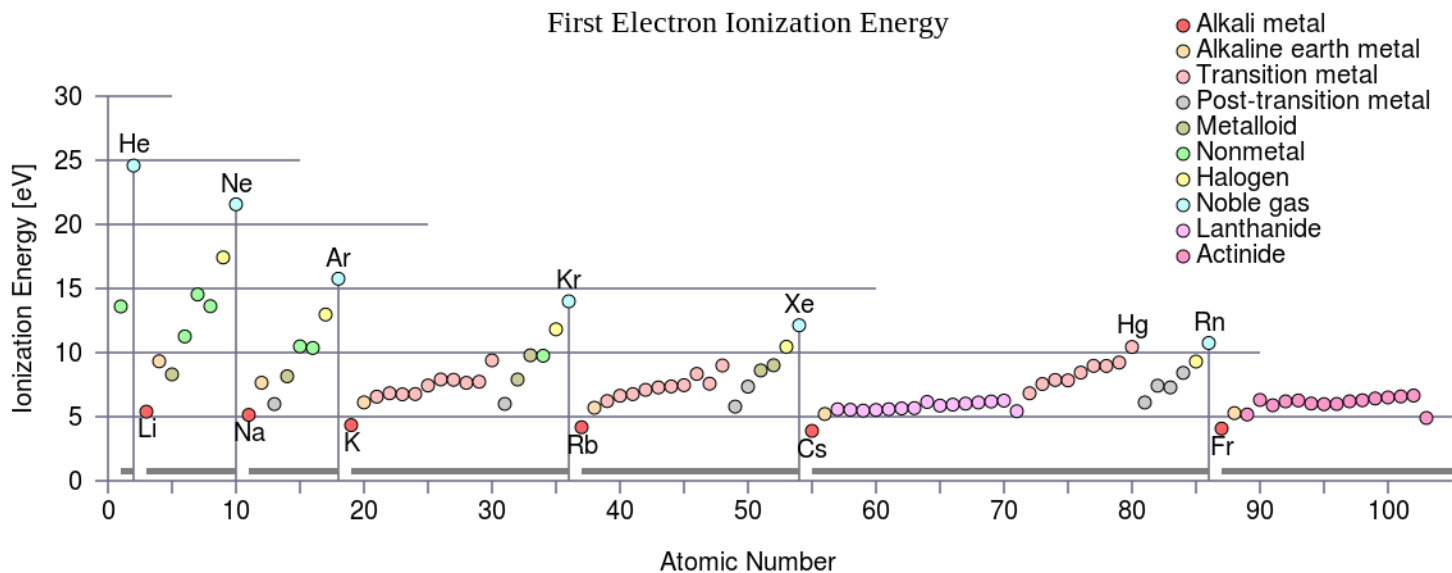
The Nobel Prize in Chemistry 1966 was awarded to was awarded to Robert S. Mulliken
"for his fundamental work concerning chemical bonds and the electronic structure of
molecules by the molecular orbital method"

<https://www.nobelprize.org/prizes/chemistry/1966/summary/>

Mulliken's Electronegativity



First Electron Ionization Energy



T A B L E 3.2

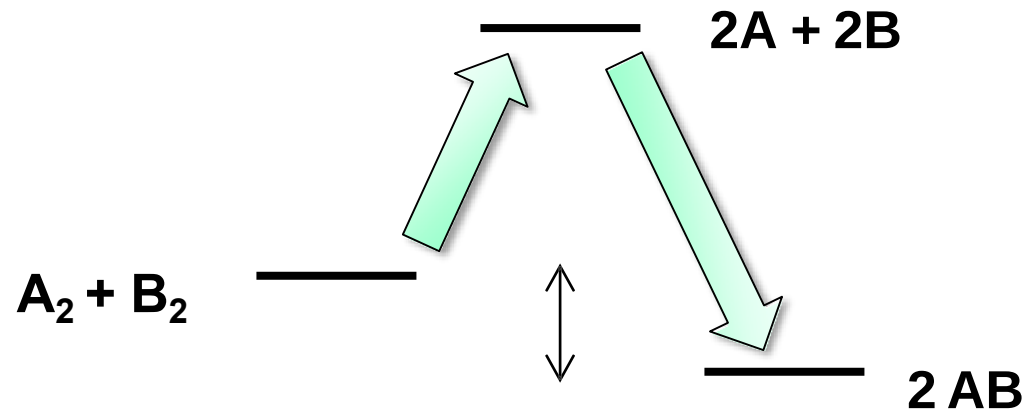
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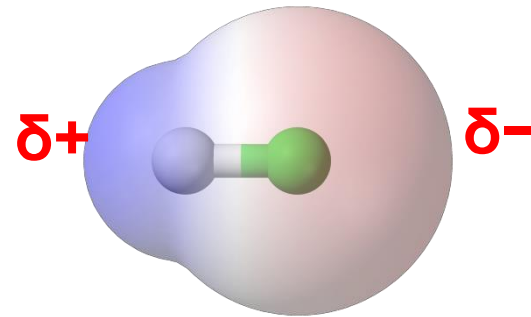
*No stable anion A⁻ exists for this element in the gas phase.

Pauling's Electronegativity

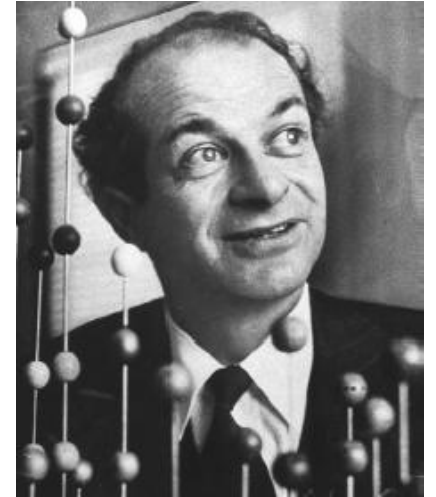
- NaCl vs. HF, HCl, HBr vs. H₂, F₂, Cl₂, Br₂



$$2 \times E_{\text{bond}} (\text{H-Cl}) > E_{\text{bond}} (\text{H-H}) + E_{\text{bond}} (\text{Cl-Cl})$$



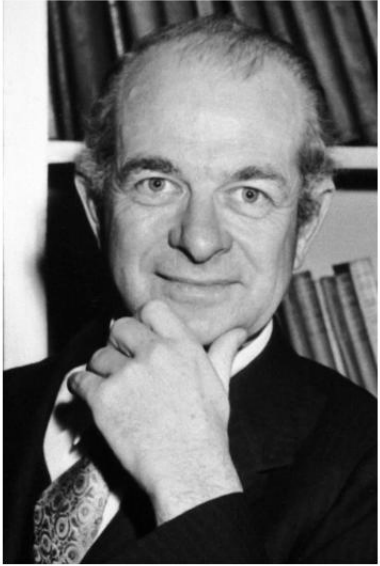
$$\Delta \propto q^2$$



Linus Pauling
(1901–1994)

The extra stability was provided by partial **charge separation** in the bond, called **ionic character**.

The Nobel Prize in Chemistry 1954



Linus Carl Pauling

Prize share: 1/1

The Nobel Prize in Chemistry 1954 was awarded to Linus Carl Pauling "for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances"

<https://www.nobelprize.org/prizes/chemistry/1954/summary/>

<https://www.nobelprize.org/prizes/peace/1962/summary/>

The Nobel Peace Prize 1962

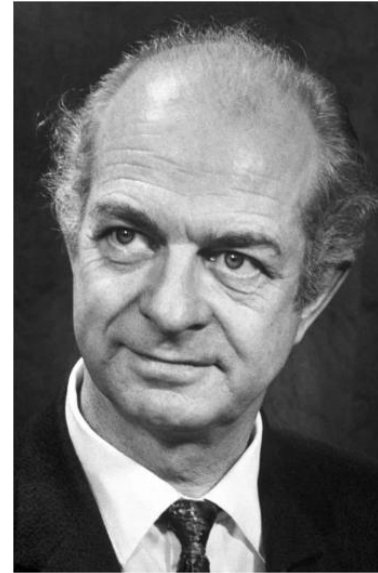


Photo from the Nobel
Foundation archive.

Linus Carl Pauling

Prize share: 1/1

The Nobel Prize in Peace 1962 was awarded to Linus Carl Pauling "for his fight against the nuclear arms race between East and West"

Pauling's Electronegativity

Pauling's Electronegativity scale: a **comparison of the bond energies** of a large number of **heteronuclear bond pairs** with those of **homonuclear diatomics** comprising the same elements.

An empirical formula:

1. Dissociation energy of A-A bond and B-B bond (both are covalent bonds): ΔE_{AA} , ΔE_{BB}
2. Dissociation energy of A-B bond (only **covalent contribution part**): $(\Delta E_{AA} \times \Delta E_{BB})^{0.5}$
3. Dissociation energy of A-B bond (**covalent contribution + ionic contribution**): ΔE_{AB}

Excess bond energy $\Delta = \Delta E_{AB} - \sqrt{\Delta E_{AA} \Delta E_{BB}}$

Pauling's definition of **electronegativity difference**

$$\chi_A - \chi_B = 0.102 \Delta^{1/2}$$

Pauling's Electronegativity

Pauling arbitrarily assigned the value 4.0 to F.

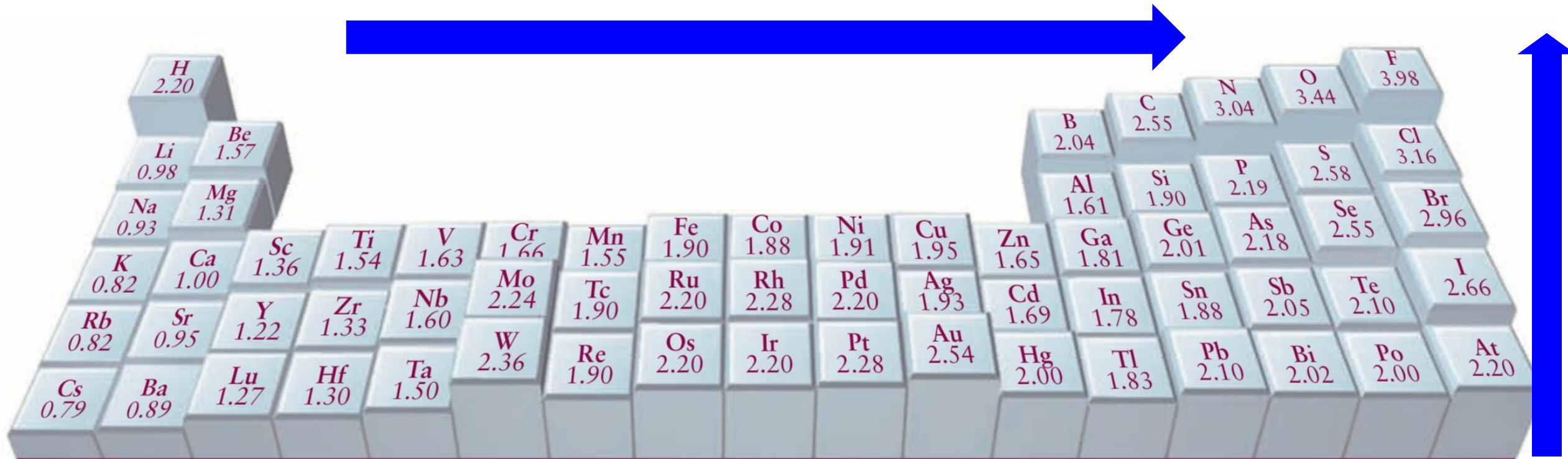


FIGURE 3.7 Average electronegativity of atoms, computed with the method that Linus Pauling developed. Electronegativity values have no units.

Without consulting tabulated values of electronegativity and guided only by the periodic table, arrange these atoms in order of increasing electronegativity: H, F, Al, O. Give a brief explanation of your answer.

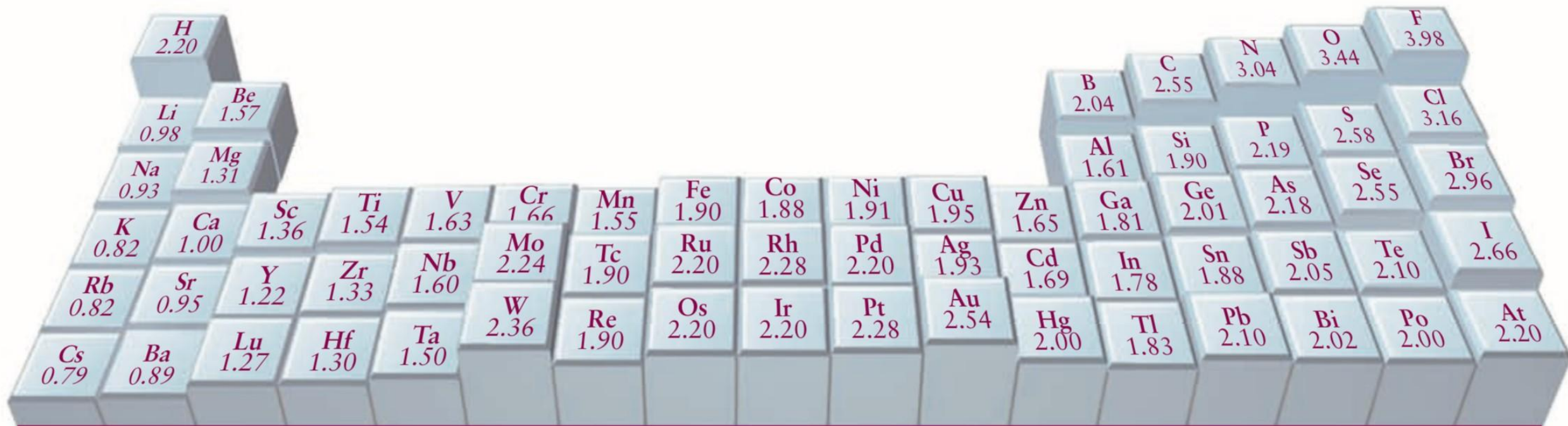


FIGURE 3.7 Average electronegativity of atoms, computed with the method that Linus Pauling developed. Electronegativity values have no units.

Homework: read “principle of modern chemistry”, chapter 3